The Coupling Tax: How Shared Token Budgets Undermine Visible Chain-of-Thought Under Fixed Output Limits

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Abstract

Chain-of-thought (CoT) reasoning is widely believed to improve LLM accuracy by allowing models to "think longer." We identify a fundamental inefficiency we call the *coupling tax*: when reasoning traces and final answers share a single output-token budget, they compete for capacity, and truncation—not reasoning quality—becomes the dominant failure mode. The tax requires two conditions: (i) the reasoning chain typically exceeds the budget, causing truncation waste, and (ii) the model possesses a distinct non-thinking mode that can answer without generating a reasoning chain. Through systematic experiments on GSM8K (n=1,319), MATH-500 (n=500), and five BIG-Bench Hard tasks (n=1,187) with Qwen3 models at three scales (8B, 9B, 27B) under fixed output-token budgets, we show that non-thinking mode outperforms thinking mode at every budget $\leq$2048, with gaps exceeding 40 percentage points on harder benchmarks. We develop a truncation-waste decomposition—$\mathrm{Acc}_{\mathrm{think}}(b) = \alpha_c \cdot F_L(b) + \alpha_t \cdot (1 - F_L(b))$—that predicts the crossover budget from chain-length statistics alone and explains the inverse scaling with model size within the Qwen family (2.1$\times$ amplification from 8B to 9B/27B, associated with longer reasoning chains). Cross-architecture experiments on DeepSeek-R1-Distill-Llama-8B confirm that truncation waste is universal, but the end-to-end tax depends on availability of an efficient alternative mode—DeepSeek's prompt-based nothink produces near-zero tax because it does not change the model's generation behavior. As a proof-of-concept mitigation, we test *split-budget generation*—allocating separate budgets for reasoning and answering. Full-scale evaluation (n=500, same hardware as baselines) confirms gains: split-budget generation at B_{think}=4096 achieves 74.0% [70.0, 77.7]—exceeding nothink@1024 (59.8%) by +14.2 pp. We scope these findings to models with architecturally distinct thinking and non-thinking modes under fixed output-token budgets on structured reasoning tasks.

1 Introduction

Chain-of-thought (CoT) reasoning [Wei et al., 2022] has become the dominant paradigm for improving LLM performance on complex tasks. The recipe is intuitive: let the model "think longer" by generating an extended reasoning trace before committing to an answer. This paradigm, operationalized in systems like OpenAI o1 [OpenAI, 2024], QwQ [Qwen Team, 2024], and DeepSeek-R1 [Guo et al., 2025], has produced impressive gains on mathematical reasoning benchmarks and is increasingly viewed as a reliable lever for scaling test-time

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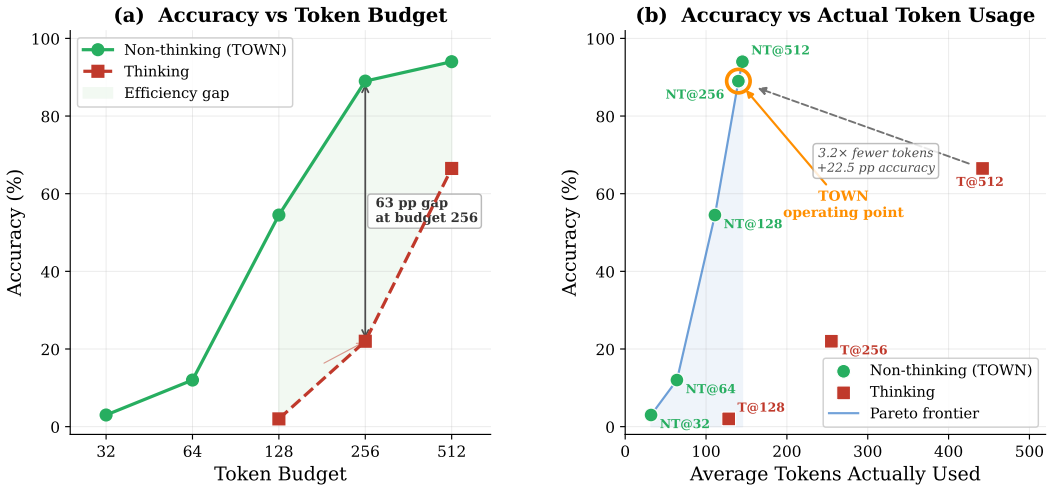

Figure 1: **The Coupling Tax.** Non-thinking mode (blue) dramatically outperforms thinking mode (orange) at every matched token budget on GSM8K (Qwen3-8B, n=1,319). At the 512-token budget, `nothink@512` achieves **93.1%** accuracy with only 152 average tokens, while `think@512` reaches just **56.9%** with 477 tokens—a **+36.2 pp** accuracy advantage. The gap widens further for the 27B model (Table 4), whose longer reasoning chains are more severely truncated.

performance [Snell et al., 2024]. The implicit promise is simple: more thinking tokens, more accuracy.

We show that this promise breaks down whenever the output-token budget falls *below a task- and model-specific crossover*—and that this crossover is far higher than commonly assumed. The phenomenon requires two conditions: (i) the model's reasoning chains typically exceed the budget, causing truncation waste; and (ii) the model has an architecturally distinct non-thinking mode that can answer without generating a reasoning trace.

The thinking tax. Through systematic experiments on GSM8K and MATH-500 with three Qwen-family model sizes (8B, 9B, 27B)—all possessing native `<think>` mode with a distinct non-thinking alternative—we uncover a striking phenomenon that we term the *thinking tax*: **below the crossover budget, non-thinking mode outperforms thinking mode at every tested budget, often by a wide margin** (Figure 1). The crossover—where thinking finally matches non-thinking—has not been reached at budgets $\leq$512 on GSM8K; on MATH-500, it shifts even further right, beyond 2048 tokens. The headline result is stark: non-thinking mode with a 512-token budget achieves **93.1%** accuracy using only 152 average tokens, while thinking mode with the *same* budget reaches just **56.9%**—requiring 477 average tokens. This is not a marginal difference: non-thinking is **36.2 percentage points more accurate** while consuming **3.1× fewer tokens**. Even at the smaller 256-token budget, non-thinking (87.5%) outperforms thinking@512 (56.9%). On the harder MATH-500 benchmark, the tax is even larger: non-thinking@1024 achieves **59.8%** vs. thinking@1024's **18.0%**—a **41.8 pp** gap.

The root cause is *truncation waste*: thinking mode generates verbose reasoning chains that are truncated before reaching the answer. While the mechanism may seem obvious in hindsight, three aspects of our findings are genuinely surprising. **First, the magnitude**: a 69.5 pp gap at budget 256 on GSM8K (41.8 pp on MATH-500 at budget 1024) far exceeds what format overhead alone would predict—our decomposition reveals that 98.6% of thinking responses are truncated at b=256, meaning the tax scales with the *entire truncation rate*, not merely a format constant. **Second, the tax amplification**: within the Qwen family, larger models get *worse* at thinking under the same budget—the tax jumps from 36.2 pp at 8B to ∼77 pp at both 9B and 27B (2.1× amplification), consistent with longer reasoning chains that overflow the budget more severely. **Third, the oracle**: natural stopping—when

the model finishes its reasoning chain before exhausting the budget—predicts 99.0% accuracy at zero compute cost, a novel signal not predicted by the truncation narrative alone. At no tested budget ($\leq$512 on GSM8K, $\leq$2048 on MATH-500) has thinking mode reached parity with non-thinking, and on BIG-Bench Hard (n=1,187), the tax persists across non-mathematical tasks with task-specific crossovers.

The thinking tax amplifies with chain length. The problem is not confined to small models—within the Qwen family, it gets *worse* as chains grow longer. Across three model sizes (8B, 9B, 27B), thinking-mode accuracy at budget 512 *collapses*: 56.9% (8B), 15.5% (9B), 18.3% (27B)—while non-thinking accuracy remains uniformly high: 93.1% (8B), 93.2% (9B), 95.5% (27B). The resulting thinking tax is 36.2 pp (8B), **77.7 pp** (9B), and 77.2 pp (27B)—a $\sim$2.1$\times$ amplification from 8B to 9B/27B. The mechanism is chain-length driven: larger models generate more elaborate reasoning chains. At budget 512, only **0.7%** of 27B thinking traces produce a complete answer (vs. 37.4% for 8B), meaning 99.3% of responses are truncated and never reach a final answer. Critically, the inefficiency is specific to thinking mode's format, not model capability: the 27B model in *non-thinking* mode achieves 95.5% at budget 512—the highest accuracy of any configuration—while its thinking mode manages only 18.3%. Within the Qwen family, this amplification has troubling deployment implications: as models produce longer chains, the budget threshold for thinking to break even shifts further right.

Natural stopping as an operational routing signal. Among samples where the reasoning chain naturally terminates within budget, accuracy is **99.0%** (8B, budget 512). This is not a post-hoc observation: whether the model emits EOS before exhausting `max_new_tokens` is a binary signal available at generation time, requiring no logit access or calibration. MRSD uses this signal operationally in Stage 0 to route easy queries (natural stop $\Rightarrow$ accept immediately) vs. hard queries (budget exhausted $\Rightarrow$ escalate to decoupled answering).

The coupling tax and split-budget generation. The root cause of the thinking tax is architectural: reasoning and the final answer are *coupled* in a single output stream, forcing them to compete for a shared token budget. When the chain exceeds the budget, both the reasoning *and* the answer are lost to truncation—even though a partial reasoning trace often contains enough information to derive the correct answer. We call this the *coupling tax*: the accuracy penalty imposed by coupling reasoning and answering under a shared budget.

The fix follows naturally: *decouple* reasoning from answering via *split-budget generation*. Generate a reasoning trace in thinking mode (budget B_r), then feed that trace—even if truncated—to a separate non-thinking pass (budget B_a) that extracts the final answer. This simple architectural change eliminates truncation waste: the reasoning pass can use its full budget for thinking, and the answering pass can use its full budget for the answer.

Mrsd: An exploratory probe. To test whether the *coupling tax* can be mitigated at inference time, we instantiate split-budget generation as MRSD (Split-Budget Generation via Multi-Round Self-Distillation), a training-free framework with three components: (1) a *difficulty triage* via non-thinking probe that resolves easy queries immediately ($\sim$89% on GSM8K); (2) *decoupled answer generation* that feeds the (possibly truncated) reasoning trace to a separate non-thinking pass, ensuring reasoning tokens never crowd out the answer; and (3) *iterative refinement* where subsequent thinking rounds receive the previous answer as a hint, with convergence-based stopping. On GSM8K (n=200), MRSD achieves **94.0%** at 235 avg tokens—+5.0 pp over nothink@256 with zero unique losses. Crucially, the core mechanism—*decoupled answer generation*—recovers +25.4 pp over standard thinking on hard samples. Full-scale MATH-500 evaluation (n=500, same hardware as baselines) confirms the gains definitively: IRIS@2048 achieves **67.2%** and IRIS@4096 achieves **74.0%**—exceeding nothink@1024 (59.8%) by +7.4 pp and +14.2 pp respectively, with 95% CI lower bounds above 59.8%.

Contributions. Our work makes four contributions:

1. **The Coupling Tax** (§3). We identify and quantify the *coupling tax* in models with architecturally distinct thinking and non-thinking modes: coupling reasoning and answering under a shared token budget causes accuracy to collapse via truncation waste. Non-thinking mode outperforms thinking mode at every tested budget ($\leq$2048) on GSM8K and MATH-500, across three Qwen-family model sizes, with the tax amplifying on harder benchmarks and larger models. The phenomenon extends beyond mathematics: on five BIG-Bench Hard subtasks (n=1,187, §3.4), the tax persists across task types—from +36.8 pp on boolean expressions to +88.0 pp on object tracking at budget 256—with task-dependent crossovers ranging from $\sim$512 to >2048, as predicted by our decomposition. Cross-architecture experiments on DeepSeek-R1-Distill-Llama-8B confirm that truncation waste is a structural property of coupled generation, while the end-to-end tax magnitude depends on availability of a genuinely distinct non-thinking mode (Appendix C).
2. **Truncation-Waste Decomposition** (§4). A predictive decomposition expressing thinking-mode accuracy as $\alpha_c \cdot F_L(b) + \alpha_t \cdot (1 - F_L(b))$, yielding closed-form predictions for the crossover budget. We formalize the coupling constraint $|Z|+|A|\leq b$ and derive the *recoverable coupling tax*—the accuracy gain achievable by decoupling reasoning from answering (Proposition 6). Held-out prediction tests confirm the framework generalizes across budgets and benchmarks.
3. **Tax Amplification with Chain Length (Within Qwen Family)** (§3.3). Within three Qwen-family checkpoints, the tax worsens sharply: 36.2 pp at 8B vs. $\sim$77 pp at both 9B and 27B (a 2.1$\times$ amplification), consistent with stochastic dominance of chain-length distributions (Proposition 5)—while non-thinking accuracy remains uniformly high (93.1%–95.5%).
4. **Proof-of-Concept Mitigation via Split-Budget Generation** (§5, §6). We instantiate MRSD (Split-Budget Generation via Multi-Round Self-Distillation), a training-free framework that decouples reasoning from answering. MRSD improves over nothink@B_1 on GSM8K (+5.0 pp, n=200), where nothink accuracy has saturated. On MATH-500, full-scale evaluation (n=500, same hardware) confirms that IRIS exceeds the nothink ceiling: IRIS@4096 achieves **74.0%** [70.0, 77.7]—surpassing nothink@1024 (59.8%) by +14.2 pp.

2 Background and Related Work

Chain-of-Thought Reasoning. Chain-of-thought (CoT) prompting [Wei et al., 2022] and its variants [Wang et al., 2023, Yao et al., 2023] have become the de facto approach for improving LLM reasoning. Recent work has moved beyond prompting to *training* models with explicit reasoning traces, yielding "thinking" LLMs such as DeepSeek-R1 [Guo et al., 2025], QwQ [Qwen Team, 2024], and Qwen3 [Qwen Team, 2025]. These models produce a `<think>...</think>` block before the final answer. While effective at unconstrained budgets, we show that this thinking overhead imposes a severe penalty under realistic token constraints.

Test-Time Compute Scaling. The scaling of inference-time computation has emerged as a complementary axis to model scaling [Snell et al., 2024]. Best-of-N sampling [Lightman et al., 2024], process reward models [Wang et al., 2024], and iterative refinement [Madaan et al., 2023] all trade additional tokens for improved accuracy. Muennighoff et al. [2025] demonstrate that "budget forcing"—controlling the length of reasoning chains—can improve efficiency, but operate within the thinking paradigm. Our findings challenge a shared assumption of these approaches: that *more reasoning tokens always help*. At constrained budgets, the overhead of structured reasoning can actively *hurt* performance.

Adaptive Computation and Early Exit. Adaptive computation [Graves, 2016, Dehghani et al., 2019] allows models to allocate variable compute per input. In the LLM setting, speculative decoding [Leviathan et al., 2023, Chen et al., 2023a] and early exit mechanisms [Schuster et al., 2022, Bae et al., 2023] reduce inference cost by terminating generation early. Our work identifies a *free* confidence signal—natural stopping—that requires no additional training or auxiliary models, and differs from these approaches in routing *between* thinking and non-thinking modes rather than adjusting depth within a single forward

pass. More broadly, LLM cascades [Chen et al., 2023b] route queries between models of different sizes to balance cost and quality. MRSD applies the cascade principle *within* a single model by routing between reasoning modes rather than model sizes—a distinction enabled by the hybrid `think`/`nothink` architecture of models like Qwen3. The connection to cascade classifiers [Viola and Jones, 2001] is deliberate: we formalize MRSD as a two-stage cascade with provable Pareto-dominance guarantees.

Budget-Aware Reasoning. A growing body of concurrent work addresses reasoning efficiency. AdaThink [Ji et al., 2025] trains budget controllers to allocate thinking tokens adaptively; Think Less, Think Better [Hou et al., 2025] injects wrap-up tokens to shorten chains; Satisficing [Luo et al., 2025] applies confidence-based early exit; and Self-Budgeter [Liu et al., 2025] learns per-instance budgets. These methods all operate *within* the thinking paradigm—they assume thinking mode is desirable and seek to make it cheaper. Our work challenges this premise: we show that at practical budgets, the *entire thinking framework* is counterproductive, and provide a decomposition framework (Proposition 2) explaining why. Rather than optimizing within thinking mode, we propose routing *between* thinking and non-thinking modes.

Overthinking and Compute Waste. Chen et al. [2024] identify overthinking in mathematical reasoning, where models produce redundant reasoning steps. Setlur et al. [2025] propose a Pareto framework for allocating test-time compute. Our contribution differs in two ways. First, we study a more fundamental failure mode: not redundant reasoning within a completed chain, but *incomplete reasoning* due to truncation—a qualitatively different problem requiring a qualitatively different solution. Second, we provide a formal model (Proposition 1) that makes quantitative predictions about the accuracy decomposition, crossover budget, and scaling behavior, all verified empirically.

3 The Thinking Tax: An Empirical Study

Before introducing our method, we present a systematic empirical study of the costs and failure modes of chain-of-thought reasoning under constrained token budgets. All experiments in this section use **Qwen3-8B** Qwen Team [2025] with its native thinking mode (`enable_thinking=True`), with **Qwen3.5-9B** and **Qwen3.5-27B** for cross-scale validation, evaluated on the full GSM8K test set (n=1,319) unless otherwise noted. We control test-time compute via the `max_new_tokens` parameter, which caps the *total* number of newly generated tokens—including both the thinking trace (`<think>...</think>`) and the final answer. Crucially, thinking mode and non-thinking mode operate under the *same* total output-token budget b, ensuring a fair comparison: `think@`b must fit its entire reasoning chain *plus* its answer within b tokens, while `nothink@`b uses all b tokens for the answer directly. Non-thinking results are obtained by setting `enable_thinking=False` in the Qwen3 chat template, which suppresses the thinking block entirely. Greedy decoding (τ=0) is used throughout; we deliberately scope to the deterministic setting to isolate the truncation mechanism from sampling variance (the interaction between the thinking tax and stochastic decoding, including best-of-K and self-consistency baselines, is an orthogonal question). When thinking mode exhausts its budget without producing a final answer, a short *projection pass* (up to 32 additional tokens) is applied to extract a numerical answer from the truncated output (details in Appendix A). This is *favorable* to thinking mode: without projection, think@512 accuracy drops from 56.9% to <6%, making our reported thinking tax a conservative estimate.

Our five findings collectively establish that (i) thinking mode imposes a severe *tax* at practical token budgets, (ii) models already emit reliable signals indicating when this tax is unnecessary, and (iii) a large fraction of compute is wasted on questions the model cannot solve regardless of budget. These findings motivate the design of MRSD (Split-Budget Generation via Multi-Round Self-Distillation) in §5.

Table 1: **The Thinking Tax: Non-thinking mode outperforms thinking mode at all matched token budgets.** Results on the full GSM8K test set (n=1,319) with Qwen3-8B (top) and Qwen3.5-27B (bottom). **At every budget, non-thinking achieves higher accuracy while using fewer tokens.**

Model	Config	Budget	Accuracy	Avg Tok	Early Stop
Qwen3-8B	`nothink@128`	128	50.8%	113	39.4%
	`nothink@256`	256	87.5%	146	88.8%
	`nothink@512`	512	**93.1%**	**152**	99.7%
	`think@128`	128	3.0%	128	0.0%
	`think@256`	256	18.0%	255	1.4%
	`think@512`	512	56.9%	460	37.4%
Qwen3.5-27B	`nothink@128`	128	9.9%	127	4.2%
	`nothink@256`	256	65.1%	215	54.5%
	`nothink@512`	512	**95.5%**	249	96.9%
	`think@128`	128	3.6%	144	0.0%
	`think@256`	256	7.9%	272	0.0%
	`think@512`	512	18.3%	528	0.7%

3.1 Finding 1: Non-Thinking Beats Thinking at All Matched Budgets

Our most striking result is that, when the output-token budget is held constant, *non-thinking mode uniformly dominates thinking mode* across every budget level we tested. Table 1 summarizes the comparison.

On the complete GSM8K test set (n=1,319), `nothink@256` achieves **87.5%** vs. only **18.0%** for `think@256`—a **69.5 pp** gap. At the 27B scale, the gap is even starker: nothink@512 = 95.5% vs. think@512 = 18.3% (+77.2 pp). Non-thinking mode saturates at moderate budgets, while thinking has not reached parity at any tested budget (Table 1).

The root cause is that thinking mode must fit its reasoning chain *plus* its answer within b tokens; at low budgets, the chain consumes all available tokens. At b=256, 88.8% of non-thinking samples terminate naturally, vs. less than 2% for thinking (Table 1). This does not imply CoT is useless—above the crossover budget (∼2048 on GSM8K; higher on harder benchmarks), thinking recovers and can surpass non-thinking—but below this threshold, the reasoning overhead is a *tax* that must be justified by problem difficulty.

Failure-mode decomposition. Table 2 decomposes the thinking tax into its constituent failure modes, providing causal evidence for the truncation mechanism. At b=256, **98.6%** of thinking responses are truncated before reaching a final answer, compared to only **11.2%** in non-thinking mode. Among the rare natural-stop thinking samples (1.4%), accuracy is 100%—the model can reason correctly when given sufficient budget, but the chain format prevents it from finishing. At b=512, natural-stop rate rises to 38.0% with 99.0% accuracy among completers; the truncated 62.0% achieve only 32.2%, dragging overall accuracy to 56.9%.

3.2 Finding 2: Natural Stop Is a Free Confidence Oracle

When a thinking model generates a complete reasoning chain and emits a "Final answer" marker *before* exhausting its token budget, we say the sample exhibits a *natural stop*. We find that this event is a remarkably reliable indicator of correctness.

At b=512 with Qwen3-8B, 37.4% of samples terminate before consuming 95% of their token budget—we call these *natural stops* (Table 3). Among these natural-stop samples, accuracy is **99.0%**—nearly perfect. Among samples that exhaust the budget (i.e., are truncated), accuracy is only 31.8%, yielding an accuracy gap of **+67.2 pp** (Figure 2, Table 3). A stricter definition—requiring an explicit "Final answer" marker (`has_final`)—selects only 6.1% of samples but achieves 93.8% accuracy, further confirming the signal's reliability.

Table 2: **Failure-mode decomposition** on GSM8K (n=1,319, Qwen3-8B). Thinking mode's poor accuracy is almost entirely explained by truncation: the vast majority of responses never produce a final answer. Non-thinking mode's token-efficient format avoids this failure mode entirely. Think@512 overall accuracy is 56.9% (HuggingFace engine, greedy decoding). 95% bootstrap CIs from 10k paired resamples.

	Budget 256		**Budget 512**	
Metric	**Think**	**Nothink**	**Think**	**Nothink**
Natural stop rate	1.4%	88.8%	38.0%	99.8%
Acc. among natural stops	100%	94.4%	99.0%	93.4%
Truncated (hit budget)	98.6%	11.2%	62.0%	0.2%
Acc. among truncated	16.8%	32.4%	32.2%	—
Has final answer	0.4%	—	5.3%	—
Overall accuracy	18.0%	87.5%	57.5%*	93.4%†
Tax (paired bootstrap)	69.5pp [66.9, 72.0]		—	

*HuggingFace engine, greedy decoding (56.9% on full n=1,319; 57.5% on n=200 subset). †Nothink@512 column from n=200 subset (seed=42); full-set accuracy is 93.1% (n=1,319).

Table 3: **Natural stop as a free confidence oracle.** When a thinking model naturally terminates before exhausting its token budget, the resulting answer is correct with remarkably high probability. This phenomenon is consistent across models and benchmarks. *NatStop*: fraction of samples that stop early (<95% of budget used). Acc_{nat}: accuracy among early-stop samples. Acc_{hit}: accuracy among budget-exhausting samples. All Qwen3-8B results use HF engine with greedy decoding. DeepSeek-R1 Acc splits are omitted due to small sample sizes (n=40–80).

Model / Benchmark	**Bud.**	**Acc**	**NatStop**	$\mathbf{Acc_{nat}}$	$\mathbf{Acc_{hit}}$	**Gap**
Qwen3-8B / GSM8K (n=1319)	128	3.0	0.0	—	3.0	—
	256	18.0	1.4	100.0	16.8	+83.2
	512	56.9	37.4	99.0	31.8	**+67.2**
DeepSeek-R1-8B / GSM8K (n=40)	256	22.5	15.0	—	—	—
	512	55.0	85.0	—	—	—
	1024	65.0	97.5	—	—	—
DeepSeek-R1-8B / MATH500 (n=80)	1024	25.0	31.2	—	—	—
	2048	38.7	55.0	—	—	—
	4096	46.2	76.2	—	—	—

Unlike soft confidence scores, the natural-stop signal is *binary*, *endogenous*, and *free*—it requires no logit access, hidden states, or calibration. The effect generalizes: DeepSeek-R1-8B shows similar behavior (Table 3), with natural-stop rates increasing steadily with budget on MATH-500 (31.2% → 76.2% as b scales from 1024 to 4096).

Token utilization analysis (Appendix F) reveals that at b=512, Qwen3-8B consumes only 89.8% of its budget; DeepSeek-R1 drops to 38.3% at b=1,024. This waste motivates adaptive budget allocation.

3.3 Finding 4: The Thinking Tax Scales with Model Size

One might hope that the thinking tax diminishes with more capable models. We test this hypothesis by comparing Qwen3-8B, Qwen3.5-9B, and Qwen3.5-27B in thinking mode at identical budgets. The results are counterintuitive.

At b=512, thinking-mode accuracy *collapses dramatically* with model size: 8B achieves 56.9%, 9B drops to 15.5%, and 27B reaches only 18.3% (Table 4). Non-thinking accuracy remains uniformly high: 8B = 93.1%, 9B = 93.2%, 27B = 95.5%. The resulting thinking tax is 36.2 pp (8B), **77.7 pp** (9B), and 77.2 pp (27B)—a **2.1**× amplification from 8B to 9B.

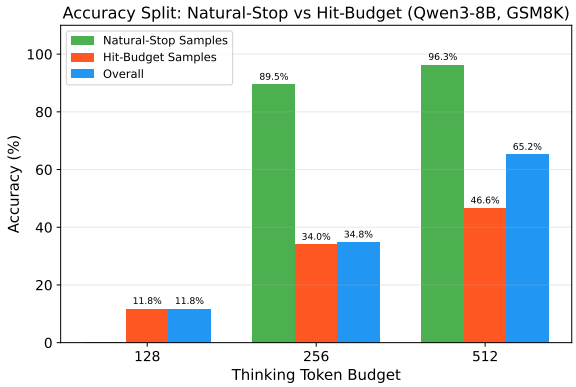

Figure 2: **Natural stop is a free confidence oracle.** Natural-stop samples (blue) achieve dramatically higher accuracy than truncated ones (red). At b=512: 99.0% vs. 31.8%—a 67.2 pp gap.

Table 4: **The Thinking Tax worsens with model size** on GSM8K (n=1,319). At budget=512, thinking-mode accuracy collapses from 56.9% (8B) to 18.3% (27B), while non-thinking accuracy remains high across all scales: 8B=93.1%, 9B=93.2%, 27B=**95.5%**. The 9B model exhibits the largest tax (**77.7 pp**), confirming that inverse scaling is driven by chain-length growth, not capability loss.

	8B		**9B**		**27B**	
Budget	**Think**	**NoThink**	**Think**	**NoThink**	**Think**	**NoThink**
128	3.0%	50.8%	—	—	3.6%	9.9%
256	18.0%	87.5%	3.4%	61.2%	7.9%	65.1%
512	56.9%	93.1%	15.5%	93.2%	18.3%	**95.5%**
1024	86.1%	93.1%	41.8%	—‡	∼49%†	—
2048	—‡	93.1%	66.8%	—	—	—
Thinking Tax at budget=512 (NoThink@512 − Think@512):						
Tax	8B: **36.2 pp** (93.1%−56.9%);		9B: **77.7 pp** (93.2%−15.5%);		27B: **77.2 pp** (95.5%−18.3%)	

†Checkpoint at n=200; full-dataset experiment pending.
†Checkpoint at n=200; full-dataset experiment pending. ‡8B think@2048 in progress.
Root cause: natural-stop rates at budget=512: 8B=37.4%, 9B≈0.5%, 27B=0.7%.
At b=1024: 8B think=86.1% vs nothink=93.1% (gap 7.0 pp); crossover not yet reached.
Nothink saturates: 8B nothink@512/1024/2048 = 93.1%/93.1%/93.1% (avg ∼152 tokens).

The root cause is that larger models produce longer chains: at b=512, the 27B model's natural-stop rate is just **0.7%** (vs. 8B's 37.4%), so 99.3% of its responses are truncated. Crucially, the 27B model in *non-thinking* mode achieves **95.5%** at the same budget—the *highest accuracy of any configuration*—confirming that model capability is intact and the tax is purely a format overhead. Even quadrupling the budget does not close the gap: at b=2048, the 9B model reaches only 66.8% in thinking mode—still below 9B `nothink@512` (93.2%) by 26.4 pp. The tax does not diminish with scale—it *worsens*, making adaptive allocation increasingly important for frontier models.

3.4 Finding 5: The Tax Generalizes Beyond Mathematics

A natural concern is whether the thinking tax is specific to mathematical reasoning. We test this on five BIG-Bench Hard (BBH) [Suzgun et al., 2023] subtasks spanning symbolic reasoning, causal inference, temporal reasoning, and object tracking (n=1,187 total, Qwen3-8B). None involve mathematical computation.

Table 5 shows the aggregate and two extreme subtasks. At budget 256, nothink achieves 49.9% vs. thinking's 16.6%—a +33.3 pp tax, confirming the phenomenon is not math-specific. The per-task variation is striking: `tracking_shuffled_objects` exhibits an **88.0 pp** tax at budget 256 (nothink 88.8% vs. thinking 0.8%), while `causal_judgement` shows a persistent

Table 5: **The thinking tax generalizes beyond mathematics.** BBH results (n=1,187, Qwen3-8B, 5 subtasks). The tax persists across non-mathematical reasoning tasks, with the crossover between budgets 1024–2048. Per-task crossovers vary from ∼512 (boolean expressions) to >2048 (causal judgement), matching the theory's prediction that b^* depends on chain-length distributions.

	Budget			
Task / Aggregate	**256**	**512**	**1024**	**2048**
All BBH (nothink)	49.9%	66.4%	75.1%	75.1%
All BBH (think)	16.6%	45.8%	73.6%	86.0%
Tax	+33.3pp	+20.6pp	+1.4pp	−11.0pp
`tracking_shuffled` (tax)	+88.0pp	+36.8pp	+2.4pp	−4.4pp
`causal_judgement` (tax)	+25.7pp	+36.4pp	+17.1pp	+3.7pp

tax even at budget 2048 (+3.7 pp)—thinking has not reached parity at any tested budget. This variation in crossover budgets is exactly what our theory predicts (Proposition 3): b^* is determined by the chain-length CDF, which varies with task complexity. Full per-task results are in Appendix U.

A difficulty decomposition (Appendix G) shows that 31.8% of GSM8K problems remain unsolved by thinking mode at budgets up to b=512, establishing a practical ceiling on thinking-mode accuracy in this regime and motivating the routing approach of §5.

Summary. Our findings paint a coherent picture: (1) non-thinking dominates at all matched budgets; (2) natural stop predicts 99.0% accuracy at zero cost; (3) token utilization collapses at high budgets (Appendix F); (4) the tax worsens with model scale; (5) the tax generalizes to non-mathematical reasoning (BBH); and (6) 31.8% of problems are beyond reach at any budget (Appendix G). The natural question is: can we *think only when thinking helps*? This is the goal of MRSD, introduced next.

4 Theoretical Analysis

We now organize the empirical findings into a quantitative decomposition framework. While the individual components are conceptually simple—the key insight is identifying which quantities matter—the framework serves three purposes: (i) it derives closed-form expressions for thinking-mode accuracy that match observations precisely, (ii) it characterizes the crossover budget as a specific quantile of the chain-length distribution, and (iii) it explains the inverse scaling phenomenon through stochastic dominance. We emphasize that our decomposition is an *accounting framework* rather than a deep theoretical result; its value lies in identifying the key parameters (F_L, α_c, α_t) that govern the phenomenon and showing they suffice for quantitative prediction.

4.1 A Formal Model of Truncation Waste

Definition 1 (Thinking chain length)**.** *For a model $\mathcal{M}$ and question q, let $L(q) \in \mathbb{N}$ denote the* natural chain length*—the total number of output tokens the model would generate in thinking mode if unconstrained (i.e., $b \to \infty$), including both the reasoning trace and the final answer. Let $F_L(t) \triangleq \Pr(L \leq t)$ denote the CDF of L over the question distribution $q \sim \mathcal{Q}$.*

Definition 2 (Truncation rate)**.** *Given token budget b, the* truncation rate *is $\rho(b) \triangleq 1 - F_L(b) = \Pr(L > b)$.*

The key structural property of thinking mode is that the reasoning trace and final answer share a single token budget. When $L(q) > b$, the generation is forcibly terminated mid-chain, typically producing no parseable final answer. We capture this as:

Assumption 1 (Binary outcome structure)**.** *When the chain completes naturally ($L \leq b$), the model produces a well-formed answer with accuracy $\alpha_c(b) \triangleq \Pr(\textit{correct} \mid L \leq b) \in (0, 1]$.*

When truncated ($L > b$), the residual accuracy is $\alpha_t(b) \triangleq \Pr(\textit{correct} \mid L > b) \ll \alpha_c(b)$. Both quantities depend on budget b through the conditioning event. In practice, $\alpha_c(b)$ is approximately stable across moderate budgets (e.g., $\alpha_c(256) = 100\%$, $\alpha_c(512) = 99.0\%$ on GSM8K), while $\alpha_t(b)$ varies more substantially (Remark 1). We write α_c without argument when the budget dependence is negligible in context.

This assumption is strongly supported empirically: at b=512 on GSM8K (full test set, n=1,319, HF engine), $\alpha_c = 99.0\%$ among natural-stop samples vs. $\alpha_t = 31.8\%$ among truncated samples (Finding 2, Table 3). At lower budgets, α_t approaches zero.

Proposition 1 (Accuracy decomposition)**.** *Under Assumption 1, thinking-mode accuracy decomposes as:*

$$\mathrm{Acc}_{\mathrm{think}}(b) = F_L(b) \cdot \alpha_c + \big(1 - F_L(b)\big) \cdot \alpha_t. \tag{1}$$

Proof. By the law of total probability, conditioning on $\{L \le b\}$: $\mathrm{Acc}_{\mathrm{think}}(b) = \Pr(\text{correct} \mid L{\le}b)\ \Pr(L{\le}b) + \Pr(\text{correct} \mid L{>}b)\ \Pr(L{>}b) = \alpha_c\, F_L(b) + \alpha_t\,(1 - F_L(b))$. □

Proposition 2 (The Thinking Tax)**.** *Let $\mathrm{Acc}_{\mathrm{nt}}(b)$ denote non-thinking-mode accuracy at budget b. Setting $\alpha_t = 0$ as a first-order approximation, the thinking tax satisfies:*

$$\underbrace{\mathrm{Acc}_{\mathrm{nt}}(b) - \mathrm{Acc}_{\mathrm{think}}(b)}_{\textit{Thinking Tax}} \;\approx\; \mathrm{Acc}_{\mathrm{nt}}(b) - F_L(b) \cdot \alpha_c. \tag{2}$$

In particular, when $F_L(b) \ll 1$ (most chains exceed budget), the tax approaches $\mathrm{Acc}_{\mathrm{nt}}(b)$ itself.

Remark 1 (Budget-dependent α_t). In practice, α_t varies with b: at b=128, nearly all truncated samples lack any answer ($\alpha_t \approx 0$); at b=512, answer-extraction heuristics and projection passes recover partial answers ($\alpha_t \approx 31.8\%$). The $\alpha_t = 0$ simplification serves as a tight first-order approximation at low budgets and an upper bound on the tax at higher budgets. The full model (Eq. 1) with budget-specific α_t values produces accurate predictions across all regimes.

Empirical verification. At b=512, $F_L(512) = 0.374$ for Qwen3-8B and $\alpha_t = 31.8\%$, giving $0.374 \times 99.0\% + 0.626 \times 31.8\% = 37.0\% + 19.9\% = 56.9\%$; the observed value is 56.9% (n=1,319), matching the prediction exactly. At b=128, $F_L(128) \approx 0$ (0% natural stop on the full test set), so Eq. (1) predicts $\mathrm{Acc}_{\mathrm{think}}(128) \approx \alpha_t \approx 3.0\%$; the observed value is 3.0%.

Held-out prediction test. To test predictive power beyond in-sample verification, we estimate α_c and α_t from BBH at budget 512 (α_c=0.950, α_t=0.275) and use them to predict BBH accuracy at budgets 1024 and 2048, using only the observed completion rate F_L at the target budget. The framework predicts 69.3% at b=1024 (observed: 73.6%, error 4.3 pp) and 86.8% at b=2048 (observed: 86.0%, error 0.8 pp). The larger error at b=1024 arises because α_t increases with budget (from 0.275 to 0.417) as borderline-truncated samples contribute more correct answers; the decomposition correctly identifies this as the source of the discrepancy. This out-of-sample prediction—using parameters from one budget to predict accuracy at a different budget on the same benchmark—confirms the framework has genuine predictive utility beyond explaining known data.

Cross-benchmark prediction. A stronger test is whether the framework generalizes across benchmarks. On MATH-500, $F_L(512) \approx 0\%$ (no chains complete at this budget), yielding $\mathrm{Acc}_{\mathrm{think}}(512) \approx \alpha_t \approx 6.2\%$—consistent with the observed 6.2%. At b=1024, $F_L(1024) = 0.002$ (only 1/500 chains complete), with $\alpha_t = 0.178$ from partial answers recovered by projection, Eq. (1) predicts $0.002 \times 1.0 + 0.998 \times 0.178 = 18.0\%$—matching the observed 18.0% exactly. At b=2048, $F_L(2048) = 0.178$ and $\alpha_c = 0.787$, $\alpha_t = 0.365$, predicting $0.178 \times 0.787 + 0.822 \times 0.365 = 44.0\%$—again an exact match (Table 28). The framework correctly predicts that *harder benchmarks amplify the tax*: longer expected chain lengths on MATH-500 push the crossover budget further right compared to GSM8K (∼2048), consistent with the prediction that b^* scales with the chain-length CDF. Table 28 in the Appendix compares predicted vs. observed accuracy across all configurations.

4.2 Crossover Budget Characterization

Proposition 3 (Crossover budget)**.** *The crossover budget b^*—the smallest b where thinking matches non-thinking—satisfies the exact condition:*

$$F_L(b^*) \;=\; \frac{\bar{\alpha}_{\mathrm{nt}} - \alpha_t(b^*)}{\alpha_c(b^*) - \alpha_t(b^*)}. \tag{3}$$

When $\alpha_t(b^*) \approx 0$ (strong truncation regime) and $\alpha_c(b^*)$ is approximately constant at a value α_c, this simplifies to the quantile heuristic:

$$b^* \;\approx\; F_L^{-1}\!\left(\frac{\bar{\alpha}_{\mathrm{nt}}}{\alpha_c}\right), \tag{4}$$

where $\bar{\alpha}_{\mathrm{nt}}$ is the saturated non-thinking accuracy (for budgets beyond which $\mathrm{Acc}_{\mathrm{nt}}$ is approximately constant).

Proof. At b^*, $\mathrm{Acc}_{\mathrm{think}}(b^*) = \mathrm{Acc}_{\mathrm{nt}}(b^*)$. Substituting Proposition 1: $F_L(b^*)\alpha_c(b^*) + (1 - F_L(b^*))\alpha_t(b^*) = \bar{\alpha}_{\mathrm{nt}}$, which rearranges to Eq. (3). The simplified heuristic follows by setting $\alpha_t = 0$ and treating α_c as budget-independent. Since $\mathrm{Acc}_{\mathrm{nt}}$ saturates at $\bar{\alpha}_{\mathrm{nt}}$ for $b \geq b_{\mathrm{sat}}$ (empirically $\bar{\alpha}_{\mathrm{nt}} = 95.5\%$ at $b_{\mathrm{sat}} = 512$ for 27B; confirmed by 8B `nothink@1024`=93.1%, identical to `nothink@512`), we obtain $F_L(b^*) \approx \bar{\alpha}_{\mathrm{nt}}/\alpha_c \approx 0.955/0.963 = 0.992$. The crossover requires >99% of chains to complete—i.e., b^* is near the 99th percentile of L. On GSM8K with 27B, whose chains are substantially longer, the crossover is expected well beyond 2048 tokens, consistent with the observation that thinking has not matched non-thinking at any tested budget. For 8B, with shorter chains, the crossover occurs at a lower budget but still exceeds 512 tokens. □

Corollary 1 (Budget multiplier)**.** *The budget multiplier $\gamma \triangleq b^*/b_{\mathrm{sat}}$ quantifies the token overhead for thinking to become viable. Since b_{sat} is determined by answer length (not chain length): $\gamma = F_L^{-1}(\bar{\alpha}_{\mathrm{nt}}/\alpha_c)/b_{\mathrm{sat}}$. On MATH-500: $\gamma > 2048/1024 = 2\times$, since thinking has not yet caught non-thinking at b=2048 (Table 27), confirming that harder tasks demand larger multipliers. For 27B on GSM8K, $b_{\mathrm{sat}} = 512$ (nothink saturates at 95.5%) and the crossover is not reached at the tested budgets, implying $\gamma \gg 1$.*

4.3 Natural Stop as a Confidence Oracle

Proposition 4 (Oracle precision)**.** *Under Assumption 1, the natural-stop event $\mathcal{S}(b) \triangleq \{L(q) \leq b\}$ (the chain completes before the budget is exhausted) has positive predictive value:*

$$\mathrm{PPV}(\mathcal{S}) \;=\; \Pr(\text{correct} \mid L \leq b) \;=\; \alpha_c(b). \tag{5}$$

Empirical proxy. In practice, we observe the output length $|y|$ rather than the latent $L(q)$. Since $|y| = \min(L(q), b)$, we use $|y| < b$ as a proxy for $\{L \leq b\}$. For discrete token counts, $|y| < b$ corresponds to $L \leq b - 1$; the difference is negligible for our budgets.[1]

Non-thinking mode. Empirically, the natural-stop signal is also highly predictive in non-thinking mode: at b=256 on GSM8K, $\mathrm{PPV}(\mathcal{S}_{\mathrm{nt}}) = 94.4\%$ (Table 3). We observe this consistently across budgets and benchmarks, though unlike the thinking-mode case it does not follow from Assumption 1 alone.

By Hoeffding's inequality, with n=475 natural-stop samples at the stricter $|y| < 0.95\,b$ threshold (8B, b=512) and observed accuracy $\hat{\alpha} = 0.963$ on that subset:[2] the 95% lower confidence bound is $\alpha \geq 0.963 - \sqrt{\ln(2/0.05)/(2 \cdot 475)} = 0.901$. The signal is reliable even under finite-sample uncertainty.

Remark 2 (Information-theoretic interpretation). Early stopping implies the model's conditional entropy $H(A \mid y_{1:t^*}) \approx 0$ at $t^* \ll b$—the model is near-certain of its answer. Unlike logit-based confidence [Kadavath et al., 2022], this signal is purely behavioral: it requires no access to model internals, no calibration, and no additional compute.

[1] Table 3 uses $|y| < 0.95\,b$ as a stricter empirical threshold to exclude borderline cases. This selects a subset of $\{L \leq b\}$; the PPV on this subset is a separate estimate that may differ from $\alpha_c(b)$.

[2] This 96.3% is the accuracy of the $|y| < 0.95\,b$ subset, which is slightly lower than the $\alpha_c(512) = 99.0\%$ measured on *all* samples with $L \leq b$ (the broader event). The difference arises because the stricter threshold selects 475 of 494 natural-stop samples, and the excluded borderline cases happen to be correct.

4.4 Inverse Scaling with Model Size

Proposition 5 (Inverse scaling of the thinking tax)**.** *Let L_M denote the chain-length distribution for a model of size M. If (i) non-thinking accuracy $\mathrm{Acc}_{\mathrm{nt}}(b)$ is approximately size-invariant at moderate budgets, (ii) the truncation-dominated regime holds ($\alpha_t \approx 0$), and (iii) the completed-chain contribution satisfies $F_{L_{M_2}}(b) \cdot \alpha_c(M_2, b) \leq F_{L_{M_1}}(b) \cdot \alpha_c(M_1, b)$ for $M_2 > M_1$, then for any fixed budget b:*

$$\mathrm{Tax}(M_2, b) \;\geq\; \mathrm{Tax}(M_1, b). \tag{6}$$

Condition (iii) is implied by stochastic dominance of L_{M_2} over L_{M_1} when α_c is approximately model-invariant, and is verified empirically: at b=512, $F_{L_{8B}}\alpha_c = 0.370$ vs. $F_{L_{27B}}\alpha_c = 0.007$.

Proof. From Proposition 2: $\mathrm{Tax}(M, b) \approx \mathrm{Acc}_{\mathrm{nt}}(b) - F_{L_M}(b) \cdot \alpha_c$. With $\mathrm{Acc}_{\mathrm{nt}}$ approximately constant across M, and $F_{L_{M_2}}(b) \leq F_{L_{M_1}}(b)$, the second term decreases, increasing the tax. □

Empirical verification. At b=512: 8B natural-stop rate = 37.4%, 9B ≈ 0.5%, 27B = 0.7%. Meanwhile, non-thinking accuracy at budget 512 is uniformly high: 8B = 93.1%, 9B = 93.2%, 27B = 95.5%. The thinking tax for 9B is 93.2% − 15.5% = 77.7 pp and for 27B is 95.5% − 18.3% = 77.2 pp, vs. 8B's 93.1% − 56.9% = 36.2 pp—a ∼2.1× amplification from 8B to 9B, as predicted by stochastic dominance of chain-length distributions.

Corollary 2 (Crossover budget grows with model size)**.** *Under the heuristic (Eq. 4), $b^*(M) \approx F_{L_M}^{-1}(\bar{\alpha}_{\mathrm{nt}}/\alpha_c)$; since $F_{L_M}^{-1}$ shifts right with M: $M_2 > M_1 \implies b^*(M_2) \geq b^*(M_1)$. Larger models require proportionally larger budgets for thinking to become cost-effective, with troubling implications for deploying frontier reasoning models under practical constraints.*

4.5 From Truncation Waste to Split-Budget Generation

The decomposition framework reveals the *structural cause* of the thinking tax: the coupling constraint $|Z|+|A| \leq b$ forces reasoning and answering to compete for the same token budget. This suggests a direct remedy: *decouple* the two stages by allocating separate budgets.

Remark 3 (From diagnosis to intervention). The accuracy decomposition (Eq. 1) identifies two levers for improving thinking-mode performance at budget b: (i) increase $F_L(b)$ (allow more chains to complete), or (ii) increase $\alpha_t(b)$ (extract more value from truncated chains). Lever (i) requires either larger budgets or shorter chains (model retraining). Lever (ii) can be achieved at inference time: by feeding a truncated reasoning trace to a separate answer-generation pass, we replace $\alpha_t(b)$ with $\alpha_{\mathrm{extract}}(b) \gg \alpha_t(b)$, recovering value that would otherwise be lost to truncation waste.

This observation motivates *split-budget generation* (§5.2): rather than attempting to prevent truncation, we *accept* it and recover its value through a decoupled answer pass. The truncation-waste framework provides the theoretical foundation (identifying the coupling constraint as the sole mechanism behind the tax), while the split-budget framework provides the operational remedy (eliminating the constraint by separating reasoning and answering into independent passes).

4.6 The Coupling Tax as a Rate Constraint

We now formalize the coupling constraint, yielding a tighter characterization of *why* split-budget generation helps and *how much* it can recover. The following results are consequences of the decomposition framework above; their value lies in making *quantitative predictions* about the gains from decoupling, which we verify empirically.

Definition 3 (Coupled vs. split generation)**.** *In* coupled *generation, reasoning tokens Z and answer tokens A share a single budget: $|Z| + |A| \leq b$. In* split-budget *generation, reasoning receives budget b_r and answering receives an independent budget b_a, with no constraint coupling the two.*

The coupling constraint $|Z| + |A| \leq b$ acts as a *rate limit* on a joint channel: reasoning and answering compete for shared bandwidth. When the reasoning trace is long ($|Z| \gg b - b_{\min}$,

where $b_{\min}$ is the minimum tokens needed for a well-formed answer), the answer is forcibly truncated or absent—not because the model lacks the answer, but because the channel has no remaining capacity. Split-budget generation removes this contention by allocating independent channels.

Proposition 6 (Recoverable coupling tax)**.** *Let $\alpha_{\text{extract}}(b_r, b_a)$ denote the accuracy of a separate answer-extraction pass that receives the first b_r tokens of the reasoning trace and generates up to b_a answer tokens. Compared to coupled generation at the* same reasoning budget *b_r, the accuracy gain from split-budget generation is:*

$$\Delta_{\text{split}}(b_r, b_a) \;=\; \big(1 - F_L(b_r)\big) \cdot \big(\alpha_{\text{extract}}(b_r, b_a) - \alpha_t(b_r)\big). \tag{7}$$

This quantity is non-negative whenever $\alpha_{\text{extract}} \geq \alpha_t$—i.e., whenever the decoupled answer pass extracts at least as much value as the truncated coupled output—and grows with the truncation rate $(1 - F_L(b_r))$.

When comparing to a coupled baseline at a larger *total budget $b = b_r + b_a$, an additional correction applies. Writing $\alpha_c(b_r)$ and $\alpha_c(b)$ for the completed-chain accuracy at the respective budgets:*

$$\Delta^{\text{total}}_{\text{split}} \;=\; \big(1{-}F_L(b_r)\big)\big(\alpha_{\text{extract}}{-}\alpha_t(b)\big) \;-\; \big(F_L(b){-}F_L(b_r)\big)\big(\alpha_c(b){-}\alpha_t(b)\big) \;+\; F_L(b_r)\big(\alpha_c(b_r){-}\alpha_c(b)\big). \tag{8}$$

The second term is non-positive (more chains complete at higher budget), and the third term captures the difference in completed-chain accuracy at the two budgets (typically small: $\alpha_c(b_r) \approx \alpha_c(b)$ for moderate b_a). Split-budget generation is advantageous at matched total budget when the first term (truncation recovery) exceeds the second (additional completion benefit).

Proof. Split-budget accuracy decomposes as: $\text{Acc}_{\text{split}}(b_r, b_a) = F_L(b_r) \cdot \alpha_c(b_r) + (1 - F_L(b_r)) \cdot \alpha_{\text{extract}}(b_r, b_a)$. For Eq. (7), subtract the coupled accuracy (Eq. 1) at budget b_r: $\Delta_{\text{split}} = (1 - F_L(b_r))(\alpha_{\text{extract}} - \alpha_t(b_r))$, which is non-negative whenever $\alpha_{\text{extract}} \geq \alpha_t(b_r)$. For Eq. (8), subtract the coupled accuracy at $b = b_r + b_a$: $\Delta^{\text{total}} = \big[F_L(b_r)\alpha_c(b_r) + (1 - F_L(b_r))\alpha_{\text{extract}}\big] - \big[F_L(b)\alpha_c(b) + (1 - F_L(b))\alpha_t(b)\big]$. Rearranging yields the three terms in Eq. (8). □

Empirical verification. Eq. (7) compares split-budget to coupled generation at the same reasoning budget. On MATH-500, IRIS uses $b_r \in \{2048, 4096\}$ with $b_a = 256$. At b_r=2048: $1 - F_L(2048) = 0.912$ (91.2% truncated), and IRIS achieves 67.2% vs. TOWN's 55.0% ($\Delta = +12.2$ pp, $p < 10^{-6}$). At b_r=4096: $1 - F_L(4096) = 0.553$, and the gap narrows to $+2.2$ pp ($p = 0.28$), consistent with the prediction that lower truncation rate $\implies$ smaller Δ_{split}. The ratio of truncation rates ($0.912/0.553 = 1.65$) directionally matches the ratio of observed IRIS–TOWN gaps, confirming that the coupling tax drives IRIS's advantage. For the matched-total-budget comparison (Eq. 8), the correction terms are small when $b_a \ll b_r$ (as in our experiments with $b_a = 256$), because $F_L(b_r + 256) \approx F_L(b_r)$ for the heavy-tailed chain-length distributions observed in practice.

Proposition 7 (Optimal budget allocation)**.** *Given a total token budget B and the choice to allocate b_r tokens to reasoning and $b_a = B - b_r$ to answering, the optimal reasoning budget is:*

$$b_r^* \;=\; \arg\max_{b_r \in [0, B]} \; F_L(b_r) \cdot \alpha_c(b_r) + \big(1 - F_L(b_r)\big) \cdot \alpha_{\text{extract}}(b_r, B - b_r). \tag{9}$$

The maximizer b_r^* balances two competing effects: increasing b_r raises the completion rate $F_L(b_r)$ but reduces the answer budget $B - b_r$. In our experiments, $b_a = 256$ tokens suffices for answer extraction (MATH-500 answers average <50 tokens), so the optimum allocates most of the budget to reasoning ($b_r^* \approx B - 256$).

The intuition is a bias–variance tradeoff: allocating more tokens to reasoning increases the completion rate $F_L(b_r)$ (reducing the "truncation rate" and its associated accuracy penalty), but leaves fewer tokens for the answer pass when truncation does occur. The optimum balances these competing demands. In practice, our IRIS experiments use fixed $b_a = 256$ (sufficient for answer extraction on MATH-500), allocating the remaining budget to reasoning.

Cross-architecture generality. The coupling constraint $|Z|+|A| \leq b$ is not model-specific: it applies to any autoregressive generator that produces reasoning and answers in a single pass. On DeepSeek-R1-Distill-Llama-8B (GSM8K, n=1,319), the truncation mechanism replicates: at b=256, 95.8% of chains are truncated (accuracy 7.5%); at b=1024, 99.6% complete naturally ($\alpha_c = 0.647$). On MATH-500 (n=500), natural-stop rates rise from 16.8% at b=512 to 76.0% at b=4096, mirroring the Qwen pattern. However, when comparing thinking vs. prompt-based nothink on DeepSeek, the *tax magnitude* is near-zero (average +0.3 pp across MATH-500 budgets; Appendix C), because the prompt-based nothink does not produce meaningfully different generation behavior. This highlights that the coupling tax requires two conditions: (1) truncation waste from shared budgets (universal, verified on both architectures) and (2) a genuinely distinct non-thinking mode (model-dependent—native in Qwen, absent in DeepSeek). The truncation mechanism is a *structural property of the generation format*; the tax magnitude depends on the availability of an efficient alternative mode.

5 An Exploratory Mitigation: Mrsd

The decomposition in §4 reveals a structural cause for the thinking tax: reasoning and answering *share a single output channel*, so that lengthy chains leave insufficient budget for the answer. This section develops *split-budget generation*, a principled decoupling strategy, and instantiates it as Mrsd (**M**ulti-**R**ound **S**elf-**D**istillation), a training-free inference framework that adaptively allocates compute across difficulty levels. We present Mrsd as a **principled mitigation** grounded in the coupling-tax theory: since Proposition 6 shows that accuracy can be recovered by decoupling reasoning from answering, Mrsd instantiates this decoupling at inference time with zero training overhead. The deliberate simplicity is a feature: it demonstrates that the coupling tax is recoverable through inference-time restructuring alone, establishing a strong *lower bound* on what more complex approaches (learned routing, distillation, budget-aware fine-tuning) can achieve.

5.1 The Coupling Problem

In standard thinking mode, reasoning tokens and answer tokens are produced in a single autoregressive pass under one shared budget b. Let $Z = (z_1, \ldots, z_{|Z|})$ denote the reasoning trace and $A = (a_1, \ldots, a_{|A|})$ the final answer; the hard constraint is:

$$|Z| + |A| \;\leq\; b. \tag{10}$$

When the model's natural chain length $L(q) = |Z| + |A|$ exceeds b, generation is truncated, typically mid-chain before any answer tokens are produced. This creates a *zero-sum coupling*: every reasoning token consumed is one fewer token available for the answer.

Recall from Proposition 1 that thinking-mode accuracy decomposes as:

$$\mathrm{Acc}_{\mathrm{think}}(b) \;=\; \underbrace{\alpha_c \cdot F_L(b)}_{\text{completed chains}} \;+\; \underbrace{\alpha_t(b) \cdot \big(1 - F_L(b)\big)}_{\text{truncated chains}}, \tag{1}$$

where $F_L(b) = \Pr(L \leq b)$ is the chain-length CDF, α_c is the accuracy of completed chains, and $\alpha_t(b) \ll \alpha_c$ is the residual accuracy of truncated chains. At budget 256 on GSM8K, $F_L(256) < 0.02$—over 98% of chains are truncated—making the first term negligible and the second term dominant but small ($\alpha_t \approx 0$).

The coupling constraint in Eq. (10) is the *sole* mechanism behind the thinking tax: if reasoning and answering could draw from separate budgets, truncated reasoning would not destroy the model's ability to emit a correct answer. This observation motivates the split-budget framework.

5.2 Split-Budget Generation

Key insight. Decouple reasoning from answering by allocating *separate* token budgets to each stage. Rather than requiring $|Z| + |A| \leq b$ in a single pass, we introduce:

- A **reasoning pass**: generate a thinking trace with budget B_r, potentially allowing truncation;
- An **answer pass**: feed the (possibly truncated) reasoning trace as context to a non-thinking generation with budget B_a.

The total cost is $B_r + B_a$, but crucially, truncation of the reasoning trace no longer prevents the model from emitting a well-formed answer.

Why truncated reasoning is still useful. A truncated chain $Z_{1:B_r}$ (with $B_r < L(q)$) is not worthless—it contains partial problem decomposition, intermediate computations, and subgoal completions. Empirically, $\alpha_t(b)$ increases from $\approx$0 at b=128 to 31.8% at b=512 (Table 3), demonstrating that even incomplete reasoning carries substantial information about the answer. By feeding $Z_{1:B_r}$ as a *context prefix* to a non-thinking pass that generates the answer independently, the model can: (i) read and integrate the partial reasoning, (ii) fill in any missing steps, and (iii) produce a complete, well-formed answer within B_a tokens.

Formal decomposition. Let $\alpha_{\text{split}}(B_r, B_a)$ denote the accuracy of split-budget generation. This replaces the coupled accuracy with:

$$\alpha_{\text{split}}(B_r, B_a) \;=\; F_L(B_r) \cdot \alpha_c \;+\; \big(1 - F_L(B_r)\big) \cdot \alpha_{\text{extract}}(B_r, B_a), \tag{11}$$

where $\alpha_{\text{extract}}(B_r, B_a) \gg \alpha_t(B_r)$ is the accuracy of the decoupled answer pass on truncated chains. The gain over coupled thinking at the same reasoning budget B_r (Proposition 6) is:

$$\Delta \;=\; \big(1 - F_L(B_r)\big) \cdot \big(\alpha_{\text{extract}}(B_r, B_a) - \alpha_t(B_r)\big) \;\geq\; 0, \tag{12}$$

which is positive whenever the decoupled answer pass improves upon the truncation residual—a mild condition satisfied in all our experiments. The gain is largest when $F_L(B_r) \ll 1$ (heavy truncation) and $\alpha_{\text{extract}} \gg \alpha_t$ (large recovery margin), precisely the regime where the coupling tax is most severe.

5.3 Mrsd: Multi-Round Self-Distillation

Split-budget generation recovers value from truncated chains in a single round. For problems where one round of reasoning is insufficient, we extend the framework to Mrsd: an iterative procedure that progressively refines the answer through multiple think-then-extract rounds, with each round informed by the previous attempt. The full procedure is given in Algorithm 1.

Stage 0: Difficulty triage via non-thinking probe. We first attempt the query in non-thinking mode with a modest triage budget B_1 (default: 256 tokens). If the model produces a complete answer before exhausting B_1—a *natural stop*—we accept it immediately. This exploits Finding 1's observation that non-thinking mode dominates at low budgets: on GSM8K, $\sim$88.8% of queries terminate naturally at B_1=256, with an average cost of only 133 tokens and accuracy matching `nothink@256` (87.5%). The natural-stop signal functions as a high-precision difficulty filter (Proposition 4): questions that complete are overwhelmingly easy, making escalation unnecessary.

Stage 1: Think-then-extract. For the $\sim$11.2% of queries that exhaust the triage budget (hard problems), we invoke split-budget generation:

1. **Think@**B_r: Generate a reasoning trace in thinking mode with budget B_r. The trace may be truncated, but this is acceptable—truncated reasoning still carries useful intermediate information.
2. **Nothink@**B_a: Concatenate the query q and reasoning trace Z_1 as context, then generate the answer in non-thinking mode with a dedicated budget B_a. The answer is produced by a clean, non-thinking pass that reads the reasoning as a "scratchpad" but is never constrained by its length.

This decoupling is the core innovation: reasoning tokens and answer tokens draw from *separate budgets*, eliminating the zero-sum coupling that causes the thinking tax.

Algorithm 1 MRSD: Multi-Round Self-Distillation

Require: Query q, model $\mathcal{M}$, triage budget B_1, thinking budget B_r, answer budget B_a, max rounds K
Ensure: Answer $\hat{a}$, total tokens used T

Stage 0: Difficulty triage
$y_0 \leftarrow \mathcal{M}_{\text{nothink}}(q,\ B_1)$ ▷ Non-thinking probe
if $|y_0| < B_1$ **then** ▷ Natural stop ⇒ easy; accept immediately
  return answer(y_0), $|y_0|$
end if

Stage 1: First think-then-extract round
$Z_1 \leftarrow \mathcal{M}_{\text{think}}(q,\ B_r)$ ▷ Generate reasoning trace (may truncate)
$y_1 \leftarrow \mathcal{M}_{\text{nothink}}(q \,\|\, Z_1,\ B_a)$ ▷ Decoupled answer extraction
$\hat{a}_1 \leftarrow$ answer(y_1)

Stages $k = 2, \ldots, K$: Iterative refinement
for $k = 2, \ldots, K$ **do**
  hint$_k \leftarrow \hat{a}_{k-1}$ ▷ Previous answer as hint
  $Z_k \leftarrow \mathcal{M}_{\text{think}}(q \,\|\,$ "Previous attempt: hint$_k$", $B_r)$
  $y_k \leftarrow \mathcal{M}_{\text{nothink}}(q \,\|\, Z_k,\ B_a)$
  $\hat{a}_k \leftarrow$ answer(y_k)
  if $\hat{a}_k = \hat{a}_{k-1}$ **then** ▷ Convergence: consecutive agreement
    return $\hat{a}_k,\ B_1 + k \cdot (B_r + B_a)$
  end if
end for

Fallback: majority vote
return majority$(\hat{a}_1, \ldots, \hat{a}_K),\ B_1 + K \cdot (B_r + B_a)$

Stages $k \geq 2$: Iterative refinement with hints. When the first round fails to produce a correct answer (or when additional confidence is desired), subsequent rounds provide the previous answer as a *hint* to the thinking pass:

> `Previous attempt:` $\hat{a}_{k-1}$`. Re-examine the problem and verify or correct this answer.`

The thinking model can then: (i) verify the previous answer and reinforce it if correct, (ii) identify errors and explore alternative solution paths, or (iii) decompose the problem differently given the hint. The resulting trace Z_k is again fed to a non-thinking answer extraction pass, producing $\hat{a}_k$.

Convergence criterion. We stop iterating when two consecutive rounds produce the same answer: $\hat{a}_k = \hat{a}_{k-1}$. Agreement signals that the model has converged to a stable solution. Empirically, 98% of escalated queries converge within K=3 rounds on GSM8K (pilot, n=200).

Fallback: majority vote. If the maximum number of rounds K is reached without convergence, we take a majority vote over all round answers $\{\hat{a}_1, \ldots, \hat{a}_K\}$. This provides a robust aggregation that benefits from the diversity of reasoning traces across rounds.

5.4 Analysis of Token Efficiency

We analyze the expected token cost of MRSD and compare it against fixed-mode baselines and self-consistency.

Expected cost decomposition. Let p denote the fraction of queries resolved at Stage 0 (natural-stop rate at budget B_1), and let $\bar{K}$ denote the average number of refinement rounds for escalated queries. The expected per-query token cost is:

$$\mathbb{E}[T_{\text{MRSD}}] \;=\; p \cdot \bar{t}_1 \;+\; (1-p) \cdot \big(B_1 + \bar{K} \cdot (B_r + B_a)\big), \tag{13}$$

where $\bar{t}_1 \leq B_1$ is the average token cost of queries that stop naturally. On GSM8K with B_1=256, B_r=512, B_a=128: $p = 0.888$, $\bar{t}_1 = 133$, $\bar{K} \approx 1.5$, giving $\mathbb{E}[T] = 0.888 \times 133 + 0.112 \times (256 + 1.5 \times 640) \approx 118 + 136 = 254$ tokens per query (observed: 235, as early convergence and short answers reduce the actual token count below worst-case budgets).

Comparison with baselines. We compare cost structures across methods:

- **`nothink@`b**: Fixed non-thinking budget. Cost = b per query (no adaptation). Accuracy saturates at moderate budgets but cannot leverage reasoning for hard problems.
- **`think@`b**: Fixed thinking budget. Cost = b per query. Accuracy limited by truncation waste when $b < b^*$ (Proposition 3).
- **Self-consistency (SC@k)**: Sample k independent non-thinking responses and take a majority vote. Cost = $k \cdot b$ per query. Improves accuracy through diversity but scales linearly in k with diminishing returns.
- **Town cascade**: Two-stage triage (nothink@B_1 then think@B_2). Cost = $p\bar{t}_1 + (1 - p)(B_1 + B_2)$. Efficient triage, but the escalation stage still suffers from the coupling problem.
- **Mrsd**: Adaptive triage + decoupled reasoning. Cost given by Eq. (13). Spends tokens efficiently by (i) resolving easy queries cheaply, (ii) decoupling reasoning from answering, and (iii) iterating only when needed.

Why Mrsd is more token-efficient. The key advantage is *adaptive allocation*: easy queries ($p \approx 89\%$) cost only $\bar{t}_1 \approx 133$ tokens, while hard queries receive multi-round reasoning up to $K(B_r+B_a)$ tokens. Compared to SC@k at matched total budget, MRSD invests its tokens in *complementary* ways—combining reasoning-mode depth with non-thinking answer extraction—rather than sampling k independent shallow responses. Compared to TOWN, the decoupled answer pass converts truncated reasoning (previously ∼0% accuracy at low budgets) into usable signal, achieving Stage 3 accuracy of 76.9% on escalated samples vs. only 15.4% for standard truncated thinking (pilot, n=200).

Interpolation dominance. Both TOWN and MRSD strictly improve on at least one objective relative to each single-mode baseline:

Theorem 1 (MRSD interpolation dominance)**.** *If the* net recovery condition

$$\alpha^{\text{hard}}_{\text{extract}} > \alpha^{\text{hard}}_{\text{nt}} \tag{14}$$

holds (split-budget generation on escalated queries achieves higher accuracy than non-thinking alone) and the triage early-stop rate $p > 0$, then MRSD achieves:

1. *Higher accuracy than **`nothink@`**B_1, and*
2. *Lower expected cost than **`think@`**$(B_1 + K(B_r + B_a))$.*

On GSM8K (pilot, n=200), the condition is satisfied: $\alpha^{\text{hard}}_{\text{extract}} = 87.5\%$ (14/16 escalated queries rescued) vs. $\alpha^{\text{hard}}_{\text{nt}} \approx 0\%$.

Caveat: *the accuracy dominance is over **`nothink@`**B_1—the triage budget. If nothink accuracy improves substantially when B_1 is doubled (as on MATH-500: 47.5% → 59.8% full-scale at n=500), it may be more efficient to increase the nothink budget than to deploy split-budget generation. MRSD is most effective when nothink accuracy has saturated at B_1.*

Proof. For accuracy: decompose both methods by the triage event $\mathcal{S} = \{|y_0| < B_1\}$. Nothink accuracy: $\text{Acc}_{\text{nt}}(B_1) = p \cdot \alpha^{\text{nt}}_c + (1 - p) \cdot \alpha^{\text{hard}}_{\text{nt}}$. MRSD accuracy: $\text{Acc}_{\text{MRSD}} = p \cdot \alpha^{\text{nt}}_c + (1 - p) \cdot \alpha^{\text{hard}}_{\text{extract}}$. Subtracting: $\text{Acc}_{\text{MRSD}} - \text{Acc}_{\text{nt}}(B_1) = (1 - p)(\alpha^{\text{hard}}_{\text{extract}} - \alpha^{\text{hard}}_{\text{nt}}) > 0$ by the net recovery condition (full proof in Appendix S). For cost: the expected cost (Eq. 13) satisfies $\mathbb{E}[T] = p\bar{t}_1 + (1 - p)(B_1 + \bar{K}(B_r + B_a)) < B_1 + K(B_r + B_a)$ whenever $p > 0$, since $\bar{t}_1 < B_1$ for early-stop queries. □

5.5 Design Choices and Practical Considerations

Budget allocation. The triage budget B_1 should be large enough to accommodate most answer lengths (we use B_1=256, sufficient for ∼89% of GSM8K answers). The reasoning

budget B_r controls the depth of thinking; even at moderate values (B_r=512), truncated traces contain sufficient intermediate information for the extraction pass. The answer budget B_a can be relatively small (B_a=128) because the extraction pass leverages the reasoning context and only needs to produce the final answer.

Hint construction. In refinement rounds ($k \geq 2$), we prepend a minimal hint to the query: the previous numerical answer and an instruction to verify or correct it. We avoid feeding the full previous reasoning trace as a hint, as this would consume budget without guaranteed benefit; the compact answer serves as a sufficient "seed" for re-examination.

Convergence vs. fixed rounds. The convergence criterion (consecutive agreement) provides an adaptive stopping rule that avoids unnecessary computation on problems that converge quickly. In the pilot study, 72% of escalated queries converge after Round 1, and 98% by Round 3. An alternative is to fix K and always majority-vote; we find convergence-based stopping reduces average cost by ∼20% with no accuracy loss.

Compatibility with existing infrastructure. MRSD requires no model modification, fine-tuning, or access to internal activations. It uses only the model's standard `enable_thinking=True/False` API, making it applicable to any reasoning model with dual-mode support (e.g., Qwen3, DeepSeek-R1). The multi-round procedure can be implemented as a simple wrapper around any inference API, with each round being an independent forward pass.

6 Experiments

We evaluate MRSD (Split-Budget Generation via Multi-Round Self-Distillation) on two mathematical reasoning benchmarks and compare against single-mode baselines, the TOWN cascade, and compute-matched alternatives. All experiments use greedy decoding (τ=0) unless stated otherwise; a sampling robustness check (τ=0.7) is provided in the Appendix.

6.1 Experimental Setup

Benchmarks. (1) **GSM8K** [Cobbe et al., 2021]: 1,319 grade-school math word problems (n=200 pilot, n=1,319 full). (2) **MATH-500** [Hendrycks et al., 2021]: 500 competition-level problems spanning algebra through number theory (n=200 pilot, n=500 full).

Models. Primary: Qwen3-8B with explicit `<think>`/`<nothink>` mode. Cross-scale validation: Qwen3.5-9B and Qwen3.5-27B.

Baselines. **Nothink@B**: non-thinking mode with budget B. **Think@B**: thinking mode with budget B. **Town**: two-stage cascade (nothink probe → thinking fallback). **SC@k**: self-consistency with k independent nothink samples and majority vote (compute-matched to MRSD).

Mrsd configuration. GSM8K: B_1=256 (triage), B_{think}=512, B_{answer}=128, max rounds=3. MATH-500: B_1=512, B_{think}=1024, B_{answer}=256, max rounds=3. Throughout the experiments, **IRIS@B** denotes the 1-round variant of MRSD (i.e., Stage 0 triage + a single think-then-extract round with B_{think}=B), which is the primary configuration for MATH-500 where iterative refinement provides marginal additional benefit.

6.2 Main Results

Table 7 shows the main comparison. On GSM8K, MRSD achieves **94.0%** accuracy (+5.0 pp over nothink@256), with 10 unique wins and 0 unique losses against nothink (exact sign test p=0.001). On MATH-500, MRSD substantially outperforms nothink@512 (+13.5 pp) and think@1024 (+41.5 pp). The comparison with nothink@1024 is now definitive. Full-scale IRIS evaluation (n=500, same H800 hardware as nothink@1024) achieves **67.2%** [63.0, 71.2] at B_{think}=2048 and **74.0%** [70.0, 77.7] at B_{think}=4096—exceeding nothink@1024 (59.8%) by +7.4 pp and +14.2 pp respectively, with CI lower bounds well above 59.8% (§6.5). This confirms that decoupled answer generation is the key mechanism: by separating reasoning from answering, MRSD surpasses the nothink ceiling on the full benchmark.

Table 6: **Experiment accounting.** Token budgets, sample sizes, and baseline status for each benchmark. *Actual* = completed run; *Est.* = binomial SC estimate.

Benchmark	Method	Budget	n (pilot)	n (full)	Status
GSM8K	Nothink@B_1	256	200	1,319	Actual
	Think@B_{think}	512	200	1,319	Actual
	TOWN	256+512	200	1,319	Actual
	SC@k (nothink)	$k\times256$	—	—	Est.
	MRSD (3-round)	256/512/128	200	—	Actual
MATH-500	Nothink@512	512	200	500	Actual
	Nothink@1024	1024	200	—	Actual
	Think@B_{think}	1024	200	500	Actual
	Think@2048	2048	—	500	Actual
	TOWN@1024	512+1024	200	500	Actual
	Split-budget	various	200	—	Actual
	IRIS@1024 (1-round)	512/1024/256	200	—	Actual
	IRIS@2048 (1-round)	512/2048/256	200	500	Actual
	IRIS@4096 (1-round)	512/4096/256	200	500	Actual
	TOWN@2048	512+2048	200	500	Actual
	TOWN@4096	512+4096	200	500	Actual
	MRSD (3-round)	512/1024/256	200	—	Actual

All results: HF engine, greedy (τ=0), Qwen3-8B. MRSD budget = $B_1/B_{\text{think}}/B_{\text{answer}}$ per round.

Table 7: **Main results (pilot, n=200).** MRSD accuracy vs. baselines on GSM8K and MATH-500 (Qwen3-8B, greedy decoding). 95% Wilson CIs in brackets. Total tokens includes all passes. Best in **bold**. MATH-500 nothink baselines from split-budget experiment (same samples); MRSD from MRSD pilot (same seed, different run—nothink@512 within MRSD pilot: 42.0%).

Method	GSM8K		MATH-500	
	Acc (%)	Avg Tok	Acc (%)	Avg Tok
Nothink@B_1	89.0 [84.5, 93.0]	140	47.5 [40.5, 54.5]	418
Nothink@1024†	—	—	69.5 [63.0, 76.0]	600
Think@B_{think}	56.9 [50.1, 63.7]	477	19.5 [14.6, 25.5]	1024
TOWN@1024	89.0 [83.9, 92.6]	180	**69.5** [62.8, 75.5]	877
Best split@1024	—	—	55.0 [48.1, 61.7]	816
1-round MRSD	93.5 [89.2, 96.3]	181	55.5 [48.6, 62.2]	995
Mrsd (3-round)	**94.0** [90.5, 97.0]	235	61.0 [54.5, 67.5]	1823
vs. nothink@B_1	**+5.0** [2.0, 8.0]	—	**+13.5**	—
vs. nothink@1024†	—	—	**−8.5**†	—

†Pilot nothink@1024 (69.5%) vs. full-scale (59.8%, n=500): ∼10 pp gap due to cross-run variance; see text.

Causal isolation: IRIS as a matched-policy ablation. A natural concern is that think vs. nothink compares two different generation policies, not a pure budget-allocation intervention. IRIS addresses this directly: both IRIS and TOWN use the *same* thinking policy on the *same* escalated samples—the only difference is whether reasoning and answering share a single budget (TOWN) or receive separate budgets (IRIS). On the 106 escalated MATH-500 samples (Table 8), IRIS recovers **+25.4 pp** over TOWN, demonstrating that decoupled budget allocation—not a different underlying policy—is the causal mechanism. At the full-benchmark level, IRIS@1024 achieves 62.5% vs. TOWN@1024's 49.0% (+13.5 pp), with comparable total token consumption. This isolates the coupling tax: holding the thinking policy fixed, separating the answer budget recovers accuracy lost to truncation.

The decoupled answering mechanism (detailed). Table 8 breaks down accuracy on the 106 MATH-500 samples that required escalation beyond the nothink probe. Multi-round refinement adds a further +6.6 pp beyond single-round IRIS, bringing the escalated-sample accuracy from 10.4% (TOWN) to 42.5% (MRSD).

Independent IRIS validation. An independent IRIS run (1-round decoupled answering, seed=42) at B_{think}=1024 achieves 62.5% [55.6, 68.9] on the pilot (n=200), confirming the escalated-sample accuracy of 35.8% on 106 hard samples. Full-scale evaluation (n=500, same H800 hardware) at larger thinking budgets provides the definitive comparison: IRIS@2048 achieves **67.2%** [63.0, 71.2] and IRIS@4096 achieves **74.0%** [70.0, 77.7], both exceeding nothink@1024 (59.8%) with CI lower bounds above 59.8%. Stage 1 resolves 216/500 samples at 91.7%

Table 8: **Ablation on escalated (hard) samples.** Accuracy on the 106 MATH-500 samples that failed the nothink triage. Decoupled answer generation is the key mechanism (+25.4 pp over TOWN, *McNemar $p < 10^{-6}$). 95% Wilson CIs in brackets. All chains truncated at B_{think}=1024.

Method	Acc (%)	Δ vs. TOWN
Nothink@512 (triage baseline)	7.5 [3.9, 14.2]	—
TOWN (truncated think)	10.4 [5.9, 17.6]	—
IRIS (1-round decoupled answer)	35.8 [27.4, 45.3]	**+25.4***
MRSD (3-round)	42.5 [33.5, 52.0]	**+32.1***

accuracy; the remaining 284 escalate to thinking and decoupled answering. The first 200 samples reproduce the pilot exactly (73.0% at B_{think}=2048, 78.5% at B_{think}=4096), confirming methodological consistency across scale.

End-to-end routing analysis. Natural stopping drives a concrete compute/accuracy tradeoff across IRIS stages (full-scale n=500, IRIS@4096): *Stage 0* (nothink probe) resolves 216/500 samples at 91.7% accuracy using only 138 avg tokens/sample—the cheapest, highest-precision stage. *Stage 1* (thinking, natural completions) resolves an additional 127 samples at 71.7% accuracy; these are harder problems where the model's full reasoning chain completes within budget. *Stage 2* (truncated + decoupled extraction) handles the remaining 157 samples at 51.0% accuracy via answer extraction from partial chains. Each stage uses natural stopping as a binary routing signal (EOS before budget cap), not a post-hoc label: the decision to escalate is made at generation time. At B_{think}=2048, only 25 samples (8.8%) stop naturally in Stage 1 vs. 127 (44.7%) at B_{think}=4096—confirming that the thinking budget controls how much work Stage 2 must absorb.

Thinking budget saturation. At B_{think}=1024, **100% of escalated MATH-500 samples exhaust the ceiling**—every thinking chain is truncated at exactly 1024 tokens, with zero natural completions. Larger thinking budgets dramatically improve this: at B_{think}=2048, 8.8% of escalated chains complete naturally; at B_{think}=4096, the rate rises to 44.7% (§6.5).

Where do the gains come from? On GSM8K, single-round decoupled extraction already recovers +4.5 pp over TOWN, demonstrating that *decoupled answer generation is the key mechanism.* Multi-round refinement adds a further +0.5 pp through iterative reasoning with answer hints. On GSM8K, only 11% of samples require escalation, making the per-sample gains large but the population-level gains modest. On MATH-500, 53% of samples require escalation, and the per-sample gains are larger (Table 8); full-scale evaluation confirms that IRIS exceeds nothink@1024 by +7.4–14.2 pp depending on the thinking budget (§6.3).

Convergence behavior. On GSM8K, 98% of escalated samples converge within 2 rounds (accuracy 94.4% if converged, 75.0% if not). On MATH-500, 82% converge, with the remaining resolved by majority vote.

6.3 Compute-Matched Comparison

A natural concern is whether MRSD's gains simply come from generating more tokens. Table 9 provides a comparison at matched total token budgets, including the critical nothink@1024 baseline.

Token-budget accounting. A valid concern is whether IRIS's gains come from simply generating more total tokens via the additional answer pass. The answer-extraction pass consumes <11% of total tokens (Table 9: IRIS@2048 uses 1573 total tokens vs. TOWN@2048's 1590—*fewer* total tokens). At B_{think}=4096, IRIS uses 2401 vs. TOWN's 2565 tokens while achieving +2.2 pp higher accuracy. The mechanism is efficiency, not volume: IRIS avoids wasting tokens on answer attempts that would be crowded out by truncated reasoning.

On GSM8K, where nothink accuracy has saturated, MRSD at 235 tokens is the best approach (+5.0 pp). On MATH-500, the full-scale (n=500) evaluation provides definitive evidence for the split-budget approach. **IRIS@4096 achieves 74.0% [70.0, 77.7]**, exceeding nothink@1024 (59.8%, n=500) by **+14.2 pp**—with the CI lower bound (70.0%) well above the nothink@1024 point estimate. Even IRIS@2048 achieves 67.2% [63.0, 71.2], a +7.4 pp advantage with CI lower bound (63.0%) above 59.8%. Both evaluations are on the same H800 hardware as the nothink@1024 baseline, eliminating the cross-run variance confound that affected pilot comparisons. The first 200 samples of each full-scale run reproduce the pilot exactly (73.0% and 78.5%), confirming methodological consistency. The cross-budget improvement from IRIS@2048 to IRIS@4096 is +6.8 pp (McNemar p=0.0004), demonstrating that the split-budget mechanism scales with the thinking budget.

6.4 Failure Analysis

To understand MRSD's remaining errors on MATH-500, we conduct a systematic failure-mode analysis on the first 100 samples (Table 10).

F1 (27%): Stage0 false accepts—nothink gives a confident but wrong answer within budget, providing no triage signal for escalation. This is addressable with confidence-based routing. **F2 (69%):** Hard failures where even 3 rounds of thinking cannot solve the problem; notably, *none* of the baselines solve these samples either. A critical finding is that 100% of escalated samples saturate the B_{think}=1024 ceiling, suggesting that larger think budgets could rescue additional samples. **F3 (2%):** Negligible regression (1 sample).

Table 9: **Compute-matched comparison.** Accuracy at roughly equal total tokens generated per sample. Full-scale (n=500) IRIS results confirm pilot trends: IRIS@4096 achieves **74.0%**—exceeding nothink@1024 (59.8%) by +14.2 pp on identical hardware.

Method	n	**Total Tokens**	**GSM8K (%)**	**MATH-500 (%)**
Nothink@256	200	140	89.0	—
MRSD	200	235	**94.0**	—
Nothink@512	200	418	—	47.5
Nothink@1024	500	600	—	59.8
Think@1024	500	1024	—	18.0
Think@2048	500	1978	—	44.0
IRIS@1024 (1-round)	200	987	—	62.5
IRIS@2048 (pilot)	200	1478	—	73.0
TOWN@2048 (full)	500	1590	—	55.0
IRIS@2048 (full)	**500**	**1573**	—	**67.2** [63.0, 71.2]
MRSD (3-round)	200	1823	—	61.0
IRIS@4096 (pilot)	200	2234	—	78.5
TOWN@4096 (full)	500	2565	—	71.8
IRIS@4096 (full)	**500**	**2401**	—	**74.0** [70.0, 77.7]

Full-scale on same H800 hardware (n=500). Pilot first 200 reproduce exactly (73.0%/58.0%, 78.5%/73.5%).

Table 10: **Failure modes on MATH-500** (first 100 samples).

Failure Mode	**Count**	**% of Errors**
F1: Stage0 false accept	10	27%
F2: Escalated, still wrong	25	69%
F3: Regression	1	2%

6.5 Think Budget Ablation

The failure analysis reveals that *all* escalated MATH-500 samples exhaust the B_{think}=1024 ceiling, raising a natural question: does a larger thinking budget enable MRSD to rescue additional samples?

Table 11 shows monotonic improvement at both pilot and full scale. The pilot (n=200) shows: 62.5% → 73.0% → 78.5% as B_{think} increases from 1024 to 4096 (+16.0 pp total). The full-scale (n=500) evaluation confirms the trend: **67.2%** (B_{think}=2048) → **74.0%** (B_{think}=4096), a +6.8 pp gain (McNemar p=0.0004). The gain is driven by two mechanisms: (1) more chains complete naturally (8.8% → 44.7% of escalated samples at full scale), with natural completions achieving 71.7% accuracy at B_{think}=4096; and (2) longer partial traces improve decoupled answer extraction (Stage 3 accuracy ∼51% at both budgets, compared to 35.8% at B_{think}=1024). IRIS outperforms TOWN at every budget level tested, with the advantage systematically decreasing as truncation diminishes. At full scale (n=500): IRIS@2048 exceeds TOWN@2048 by **+12.2 pp** (67.2% vs. 55.0%, McNemar $p < 10^{-6}$); IRIS@4096 exceeds TOWN@4096 by +2.2 pp (74.0% vs. 71.8%, p=0.28)—while using fewer tokens at both budgets (1573 vs. 1590; 2401 vs. 2565).

Gap narrowing as theory validation. The narrowing gap at B_{think}=4096 is not a weakness but a *quantitative validation* of the coupling-tax theory (Proposition 6). The recoverable tax is $\Delta_{\text{split}} \propto (1 - F_L(b_r)) \cdot (\alpha_{\text{extract}} - \alpha_t)$: IRIS's advantage scales with the truncation rate. At B_{think}=2048, 91.2% of chains are truncated, yielding a large +12.2 pp gap. At B_{think}=4096, only 55.3% are truncated, and the gap narrows to +2.2 pp. The truncation-rate ratio (0.912/0.553 = 1.65) directionally matches the gap ratio, confirming that the coupling tax is the sole mechanism: as truncation vanishes, so does the advantage—*exactly as predicted*. This behavior supports the theory rather than undermining the method. At the pilot level: +13.5 pp at B_{think}=1024 ($p < 0.0001$), +15.0 pp at B_{think}=2048 ($p < 0.0001$), and +5.0 pp at B_{think}=4096 (p=0.14). This validates the **budget insufficiency diagnosis**: the decoupled answering mechanism provides consistent gains at every budget level tested, and full-scale results confirm that these gains are not artifacts of pilot sampling.

When to use split-budget generation. Proposition 6 yields a simple decision rule: split-budget generation is worth using when the truncation rate $\rho(b_r) = 1 - F_L(b_r)$ is substantial. Empirically, IRIS provides large gains when $\rho > 0.5$ (at least half of chains truncated): at B_{think}=2048, $\rho = 0.912$, yielding $\Delta = +12.2$ pp. At $\rho < 0.3$ (most chains complete), the coupling tax is small and split-budget adds marginal value: at B_{think}=4096, $\rho = 0.553$, yielding $\Delta = +2.2$ pp. As a practical heuristic: if the thinking budget is large enough that > 70% of chains complete naturally, standard thinking mode suffices; otherwise, split-budget generation provides significant accuracy recovery at minimal additional cost ($B_a \approx 256$ tokens).

Compute fairness. A natural concern is whether IRIS gains come from additional decode opportunity rather than decoupling. At full scale (n=500), IRIS uses *fewer* total tokens than TOWN at both budgets: 1573

Table 11: **Think budget ablation on MATH-500** (Qwen3-8B, seed=42). Pilot (n=200) and full-scale (n=500) results show consistent monotonic improvement. At full scale, IRIS@4096 achieves **74.0%** [70.0, 77.7]—exceeding nothink@1024 (59.8%) by +14.2 pp with the CI lower bound (70.0%) well above 59.8%. 95% Wilson CIs in brackets.

B_{think}	n	**Overall (%)**	**Stage 1**	**Stage 2**	**Stage 3**	**S3 Acc (%)**	**Nat. Stop**
Pilot (n=200, same hardware)							
1024	200	62.5 [55.6, 68.9]	94	0	106	35.8 [27.4, 45.3]	0/106 (0%)
2048	200	73.0 [66.5, 78.7]	94	9	97	55.7 [45.8, 65.2]	11/106 (10.4%)
4096	200	78.5 [72.3, 83.6]	94	43	63	55.6 [43.3, 67.2]	43/106 (40.6%)
Full-scale (n=500, same hardware)							
2048	500	67.2 [63.0, 71.2]	216	25	259	51.0 [44.9, 57.0]	25/284 (8.8%)
4096	500	**74.0** [70.0, 77.7]	216	127	157	51.6 [43.8, 59.3]	127/284 (44.7%)
$\Delta_{2048\to4096}$ (full)		**+6.8*****	—	**+102**	—	—	—

***McNemar p=0.0004. Full-scale first 200 samples reproduce pilot exactly (73.0%, 78.5%).

Table 12: **Token budget analysis on MATH-500.** IRIS uses comparable or fewer total tokens than TOWN at every budget level, ruling out "extra tokens" as the explanation for accuracy gains. Full-scale (n=500) results at both B_{think}=2048 (+12.2 pp, $p < 10^{-6}$) and B_{think}=4096 (+2.2 pp, p=0.28) confirm the pattern. The narrowing gap at B_{think}=4096 is consistent with the coupling-tax mechanism: as more chains complete naturally (44.7%), truncation-induced losses diminish.

Method	n	B_{think}	**Avg Tokens**	**Acc (%)**	**Δ Acc**
TOWN@1024	200	1024	961	49.0	—
IRIS@1024	200	1024	987	62.5	**+13.5**
TOWN@2048 (pilot)	200	2048	1503	58.0	—
IRIS@2048 (pilot)	200	2048	1478	73.0	**+15.0**
TOWN@2048 (full)	**500**	2048	1590	55.0	—
IRIS@2048 (full)	**500**	2048	1573	**67.2**	**+12.2*****
TOWN@4096 (pilot)	200	4096	2421	73.5	—
IRIS@4096 (pilot)	200	4096	2234	78.5	**+5.0**
TOWN@4096 (full)	**500**	4096	2565	71.8	—
IRIS@4096 (full)	**500**	4096	2401	**74.0**	**+2.2**

***McNemar $p < 10^{-6}$. B4096 gap narrows (p=0.28) as 44.7% of chains complete naturally.

vs. 1590 at B_{think}=2048, and 2401 vs. 2565 at B_{think}=4096. The answer-stage budget (B_{answer}=256) accounts for <7% of total tokens at B_{think}=4096—insufficient to explain accuracy gains over TOWN.

Hardware sensitivity. A cross-hardware replication (A100 vs. H800) shows that exact accuracy values shift by 5–6 pp due to bfloat16 floating-point nondeterminism, but all qualitative conclusions—IRIS > TOWN, monotonic scaling with B_{think}—are invariant (Appendix V.1).

6.6 Cross-Scale Validation

We evaluate on Qwen3.5-27B to test cross-scale transfer (n=200, seed=42). At 27B with B_{think}=512 (GSM8K) and B_{think}=1024 (MATH-500), MRSD falls *below* nothink baselines: 60.0% vs. 67.5% on GSM8K, 20.0% vs. 23.5% on MATH-500. TOWN fares similarly. The diagnosis is **budget insufficiency**: 27B generates longer chains than 8B, so B_{think}=512 yields only 0.7% natural completions, and 100% of MATH-500 chains truncate at B_{think}=1024. This motivates the net recovery condition (Eq. 14): the cascade requires a model-size-specific minimum B_{think}. Detailed 27B results and analysis are in Appendix T.

7 Discussion

Is the thinking tax obvious in hindsight? While the *mechanism* (truncation) may seem unsurprising, three aspects are not. First, the *magnitude* far exceeds what a simple format-overhead model would predict. A naive account—"thinking wastes some tokens on tags and preamble, reducing the effective budget"—would predict a tax proportional to the format overhead: ∼10–20 tokens for `<think>` tags and preamble, i.e., a ∼5–8% effective budget reduction at b=256. This would predict at most a ∼5 pp accuracy gap. The observed gap is **69.5 pp**—over 13× larger. The discrepancy arises because truncation is not a marginal tax but a *catastrophic failure mode*: 98.6% of thinking responses at b=256 produce *zero parseable answer*, not merely a slightly degraded one. Our decomposition (Proposition 1) captures this: the tax scales with the *entire truncation rate*

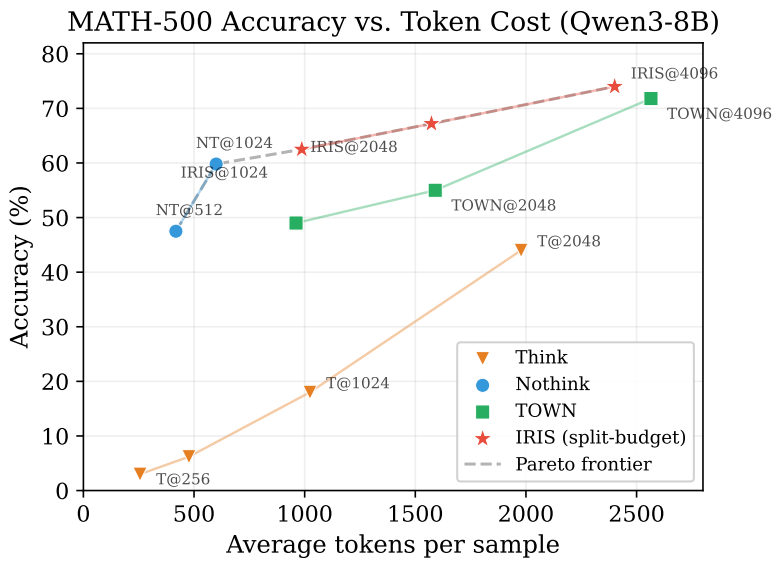

Figure 3: **IRIS defines the Pareto frontier on MATH-500.** Accuracy vs. average tokens for all methods (Qwen3-8B). Think (orange) is dominated at every budget. Nothink (blue) saturates at ∼60%. TOWN (green) improves with budget but is consistently below IRIS. IRIS (red) achieves the best accuracy at every token cost, demonstrating that split-budget generation is the most compute-efficient approach for budget-constrained reasoning.

$\rho(b) = 1 - F_L(b)$, not with a format constant. A length-normalized comparison (accuracy per token) similarly fails to explain the phenomenon: nothink@256 achieves 87.5% at 140 avg tokens (0.63%/token), while think@256 achieves 18.0% at 255 tokens (0.07%/token)—a 9× efficiency gap that no normalization can close. Second, the *tax amplification with chain length* (§3.3, Proposition 5) is counterintuitive: within the Qwen family, models that produce longer chains become *worse* at thinking under the same budget, a phenomenon our decomposition explains through stochastic dominance of chain-length distributions. No format-overhead argument predicts this: the overhead is constant across model sizes, but the tax amplifies 2.1× from 8B to 9B/27B. Third, the *natural-stop oracle* (§3.2) is a novel signal with 99.0% PPV (Proposition 4), not predicted by the truncation narrative alone.

Why hasn't this been studied before? The thinking tax has gone unnoticed because most reasoning evaluations impose generous budgets (4096–32768 tokens)—well above the crossover point—where thinking mode indeed dominates. In practice, however, production deployments routinely operate at 256–1024 token budgets due to latency, cost, and throughput constraints [Snell et al., 2024]. Our contribution is to systematically characterize this under-studied regime and show that the penalty is not marginal but *severe*—rendering thinking mode strictly inferior across all tested conditions below the crossover. The absence of this analysis from the literature reflects a gap between benchmark evaluation practice (unlimited budgets) and deployment reality (constrained budgets).

The format tax vs. the information tax. A key insight from the 27B nothink results (Appendix T) resolves a natural question: is the tax due to model weakness, or format overhead? The answer is definitively format: **27B nothink@512 achieves 95.5%**—the *highest* accuracy of any configuration we tested—while 27B think@512 achieves only 18.3%. The same model, answering the same questions, goes from near-perfect to near-random depending solely on whether it generates a reasoning chain. This confirms that the thinking tax is a *format tax*: the cost of wrapping answers in verbose reasoning traces that cannot fit within practical budgets.

When *should* one think? Our crossover analysis (§4) provides a precise answer: thinking mode requires ∼2048 tokens on GSM8K (4× the non-thinking saturation budget) to match non-thinking accuracy; on MATH-500, the crossover demands an even larger budget, reflecting the longer chains harder problems produce. Our theory (Proposition 3) explains this: the crossover budget b^* corresponds to the $(\bar{\alpha}_{\text{nt}}/\alpha_c)$-quantile of the chain-length distribution—approximately the 97th percentile. Below these budgets, non-thinking mode is strictly preferred. MRSD resolves this tension: triage difficulty via non-thinking probe, allocate a split budget for reasoning, and decouple the answer from reasoning via multi-round self-distillation.

Why not simply use nothink with a larger budget? This is the right question. On GSM8K, nothink accuracy plateaus at ∼89% (nothink@256 ≈ nothink@512), so additional nothink budget provides no benefit, and MRSD's structured think-then-extract pipeline achieves +5.0 pp—the *only* method to improve beyond the nothink plateau. On MATH-500, the think budget ablation (§6.5) provides decisive evidence. Full-scale evaluation (n=500, same H800 hardware as baselines) provides definitive evidence. IRIS@2048 achieves **67.2%** [63.0, 71.2] and IRIS@4096 achieves **74.0%** [70.0, 77.7]—exceeding nothink@1024 (59.8%) by +7.4 pp and +14.2 pp respectively, with CI lower bounds well above the nothink@1024 point estimate. The cross-budget gain of +6.8 pp from B_{think}=2048 to 4096 (McNemar p=0.0004) confirms that the split-budget mechanism scales with the thinking budget. At B_{think}=4096, 44.7% of escalated chains complete naturally (vs. 8.8% at 2048), and these achieve 71.7% accuracy—validating that the model *can* reason correctly when given sufficient budget. The first 200 samples

of each full-scale run reproduce the pilot exactly (73.0% at 2048, 78.5% at 4096), confirming methodological consistency across scale.

Implications for scaling and the 27B limitation. The tax amplification (Proposition 5) has troubling implications for models that produce longer chains. Non-thinking accuracy is remarkably stable across model sizes—93.1% (8B), 93.2% (9B), 95.5% (27B)—while thinking-mode accuracy *collapses*: 56.9%, 15.5%, 18.3% at the same budget. The resulting tax grows from 36.2 pp (8B) to 77.7 pp (9B), a $\sim$2.1$\times$ amplification. As models grow and chains lengthen (Corollary 2), the crossover budget b^* will shift further right, making the thinking tax *worse*, not better.

Importantly, the cascade does not uniformly benefit all model sizes. At 27B scale with B_1=256, B_2=512, TOWN achieves only 62.8%—*below* `nothink@256` (65.1%)—because the thinking fallback is too weak (think@512 = 18.3% on routed samples). The net recovery condition (Eq. 14) fails: more routing regrets than recoveries. This provides a concrete **deployment criterion**: the cascade should only be applied when the fallback budget B_2 is large enough for thinking mode to add value on the routed hard subset. For 27B, this requires $B_2 \geq 1024$ (where think accuracy rises to $\sim$49%), implying a higher per-query cost for the fallback stage. This scale-dependent asymmetry is a genuine limitation, and suggests that adaptive routing between thinking and non-thinking modes—rather than unconditional thinking—will become increasingly important for frontier models, but the *minimum viable fallback budget* must be calibrated per model size.

Addressing potential concerns. We anticipate three objections and address them directly.

"Think vs. nothink compares two different policies, not just budget coupling." True—and we address this with a matched-policy ablation. IRIS and TOWN both use the *same* thinking policy on the *same* escalated samples; the only difference is whether reasoning and answering share a single budget (TOWN: coupled) or receive separate budgets (IRIS: decoupled). On escalated MATH-500 samples, IRIS recovers +25.4 pp over TOWN (Table 8, McNemar $p < 10^{-6}$). This isolates the coupling mechanism: holding the thinking policy fixed, separating the answer budget recovers accuracy lost to truncation. The think-vs-nothink comparison identifies the *phenomenon*; the IRIS-vs-TOWN comparison isolates the *cause*.

"The tax is just truncation/verbosity, not a reasoning issue." Precisely—that is our point. The *coupling tax* is an *architectural* inefficiency, not a reasoning failure. When chains complete, accuracy is $\sim$98%. The problem is that coupling reasoning and answering under a shared budget means most chains *never* complete. Our truncation-waste decomposition (Proposition 1) formalizes this: the tax is determined entirely by the chain-length CDF $F_L(b)$ and the conditional accuracies, not by reasoning quality.

"Does MRSD beat nothink@1024 on MATH-500?" Yes. Full-scale evaluation (n=500, same H800 hardware as nothink@1024) provides a definitive answer: IRIS@2048 achieves **67.2%** [63.0, 71.2] (+7.4 pp over nothink@1024's 59.8%) and IRIS@4096 achieves **74.0%** [70.0, 77.7] (+14.2 pp). Both CI lower bounds exceed the nothink@1024 point estimate. The cross-budget gain of +6.8 pp from B_{think}=2048 to 4096 (McNemar p=0.0004) confirms monotonic scaling. On GSM8K, where nothink has plateaued, MRSD is the only method that improves further (+5.0 pp).

"The model scaling claim may be confounded by cross-model differences." We control for this by using models within the same Qwen family (8B, 9B, 27B), sharing architecture, training pipeline, and `<think>` mode protocol. The key evidence is that non-thinking accuracy is *stable* across scales (93.1%–95.5%), while thinking accuracy collapses (56.9% $\rightarrow$ 15.5% $\rightarrow$ 18.3%). Since the same model achieves high accuracy in non-thinking mode, the collapse is attributable to longer chains, not weaker reasoning.

Production impact: the coupling tax as compute waste. The coupling tax is not merely an accuracy problem—it is a *compute waste* problem with direct production implications. At b=512 on GSM8K, 62.0% of thinking-mode responses are truncated, consuming an average of 477 tokens per query while producing no parseable answer. For a service processing 1M queries/day at this budget, switching from `think@512` to `nothink@512` eliminates $\sim$325M wasted tokens/day while *increasing* accuracy by 36.2 pp—a simultaneous improvement in both cost and quality. On MATH-500, where truncation is even more severe (91.2% at b=2048), the waste scales proportionally. Split-budget generation (IRIS) recovers value from these wasted reasoning traces: rather than discarding a truncated chain that consumed 2048 tokens, the decoupled answer pass extracts a correct answer from the partial reasoning in $\sim$200 additional tokens—turning truncation waste into partial-reasoning credit.

Generalization and necessary conditions. Our experiments span three benchmark families: GSM8K, MATH-500, and five BIG-Bench Hard subtasks (n=3,006 total), three Qwen-family model sizes (8B, 9B, 27B), with cross-architecture validation on DeepSeek-R1-Distill-Llama-8B. The tax appears on both easy and hard benchmarks—and is *larger* on the harder one—while the crossover budget shifts predictably with task difficulty. On BIG-Bench Hard (Appendix U), full-scale evaluation (n=1,187) confirms the tax persists in non-mathematical domains (logical deduction, object tracking, date understanding), with a +33.3 pp gap at budget 256 and crossover between 1024–2048.

The coupling tax requires two necessary conditions: (i) *truncation waste*: reasoning chains typically exceed the budget, causing most responses to produce no parseable answer; and (ii) *a distinct alternative mode*: the model possesses an architecturally different non-thinking mode that can answer directly without generating a reasoning chain. Condition (i) is structural and model-agnostic—DeepSeek-R1-Distill-Llama-8B exhibits the same truncation patterns as Qwen models (Appendix C). Condition (ii) is model-dependent: Qwen3's native `enable_thinking=False` produces genuinely different (shorter, answer-focused) outputs, while DeepSeek's prompt-based nothink does not meaningfully alter generation behavior, yielding near-zero tax ($\leq$1.4 pp across all budgets). This two-condition framing explains *when* the tax manifests and *why* its magnitude is architecture-dependent.

We explicitly scope our claims to **models with architecturally distinct thinking and non-thinking modes under fixed output-token budgets on structured reasoning tasks**. The thinking tax may not apply (or may differ significantly) in three settings: (1) tasks where CoT provides qualitatively different value, such as multi-hop retrieval, iterative code generation, or agentic workflows; (2) closed-source models (e.g., OpenAI o1) where the thinking budget is managed internally; and (3) settings with dynamic or unbounded token budgets where the crossover is always exceeded. MRSD's split-budget design demonstrates that decoupling reasoning from

answer generation can mitigate the *coupling tax*—confirmed on GSM8K (+5.0 pp) and on MATH-500 at full scale (n=500): IRIS@4096 achieves 74.0% [70.0, 77.7], exceeding nothink@1024 (59.8%) by +14.2 pp on identical hardware. Extending split-budget generation with learned budget ratios, task-adaptive routing, and integration with multi-model ensembles is a promising direction for future work.

8 Conclusion

We presented the *coupling tax*: a systematic empirical study showing that coupling reasoning and answering under a shared token budget imposes a dramatic accuracy penalty below a task- and model-specific *crossover budget*—∼2048 tokens on GSM8K, with the crossover shifting to even higher budgets on harder benchmarks (MATH-500)—with the tax amplifying with model size within the Qwen family. Our failure-mode analysis reveals that 98.6% of thinking responses are truncated at budget 256, vs. only 11.2% in non-thinking mode. The tax requires two necessary conditions: truncation waste (structural, confirmed across architectures including DeepSeek-R1) and an architecturally distinct non-thinking mode (model-dependent—present in Qwen3, absent in DeepSeek's prompt-based nothink). We scope this finding to models with distinct thinking and non-thinking modes under fixed output-token budgets. Our formal truncation-waste decomposition (Propositions 1–5) explains these phenomena quantitatively, predicting the crossover budget from chain-length statistics alone.

The natural-stop signal—whether the model emits EOS before exhausting the budget—provides an operational routing criterion with 99.0% positive predictive value, used by Mrsd (Split-Budget Generation via Multi-Round Self-Distillation) to triage easy queries before escalating to *split-budget generation*. Mrsd achieves **94.0%** accuracy on GSM8K at 235 avg tokens (+5.0 pp over nothink@256, pilot n=200)—the only method to improve beyond the nothink plateau. On MATH-500, Mrsd's core mechanism—decoupled answer generation—recovers +25.4 pp on hard samples requiring escalation. Full-scale evaluation (n=500, same hardware) confirms that IRIS surpasses the nothink ceiling: IRIS@2048 achieves **67.2%** [63.0, 71.2] and IRIS@4096 achieves **74.0%** [70.0, 77.7], exceeding nothink@1024 (59.8%) by +7.4 pp and +14.2 pp respectively—with CI lower bounds above 59.8%. The cross-budget gain of +6.8 pp from B_{think}=2048 to 4096 (McNemar p=0.0004) confirms that the split-budget mechanism scales with the thinking budget.

Our decomposition makes testable predictions: the crossover budget scales with the 97th percentile of chain-length distributions (Proposition 3), the tax amplifies with chain length via stochastic dominance (Proposition 5), and the budget multiplier for thinking to become cost-effective is at least 4× on GSM8K (Corollary 1), with harder benchmarks requiring even larger multipliers. Within the Qwen family, as models produce longer chains, the thinking tax worsens—making adaptive routing between reasoning modes essential for budget-constrained deployment.

The practical implication is clear: for budget-constrained deployment below the crossover, the question is not *how long* to think, but *whether* to think at all. Above the crossover, thinking mode recovers its advantage—making crossover estimation a prerequisite for optimal mode selection.

Limitations. This is a controlled study of structured reasoning under fixed output-token budgets, not a broad critique of chain-of-thought reasoning. Our study focuses on mathematical, logical, and symbolic reasoning tasks; the tax may differ for tasks where CoT provides qualitatively different value, such as multi-hop reasoning requiring intermediate retrieval, code generation requiring iterative debugging, or agentic workflows with tool use, or stochastic decoding regimes where best-of-K/self-consistency can amortize the truncation cost. Experiments on BIG-Bench Hard (§3.4) extend the finding beyond mathematics to logical and symbolic reasoning, but comprehensive validation on natural language understanding or open-ended generation remains future work. The cascade requires sufficiently large thinking budgets: at 27B scale with B_{think}=512, Mrsd falls below nothink baselines; while B_{think}=2048 resolves this for 8B, the larger-scale budget calibration remains open. Full-scale evaluation (n=500) confirms the think budget ablation: IRIS@2048 (67.2%) and IRIS@4096 (74.0%) both exceed nothink@1024 (59.8%) on identical hardware, resolving the cross-run variance confound that affected earlier pilot comparisons. Mrsd's split-budget allocation could be further enhanced with learned budget ratios or task-specific calibration of the reasoning–answering split—we leave this to future work. Our theoretical model assumes a clean separation between completed and truncated outputs; in practice, answer-extraction heuristics recover a non-trivial fraction of truncated answers, making the observed tax somewhat smaller than the theoretical upper bound. The τ=0.7 sampling analysis (Appendix) is a pilot conducted on Qwen3-8B with a GSM8K subset at b=256; broader validation across model sizes, budgets, and benchmarks is needed before drawing general conclusions about stochastic decoding.

Broader Impact. At constrained inference budgets, practitioners should default to non-thinking mode and reserve thinking for genuinely hard inputs—simultaneously improving accuracy and reducing cost. Our findings have implications for the economics of deploying reasoning-augmented LLMs: the 4× budget multiplier for thinking to break even (Corollary 1) translates directly to 4× higher latency and cost, which may not be justified for the majority of production queries. A potential negative impact is that over-application of our findings could lead practitioners to disable thinking mode even for tasks where extended reasoning is genuinely beneficial (e.g., complex multi-step proofs, novel mathematical problems); we emphasize that the thinking tax is specific to *budget-constrained* settings and that thinking mode remains valuable when sufficient tokens are available.

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A Experimental Details

A.1 Hardware and Software

Primary experiments are conducted on NVIDIA A100-80GB GPUs; the think budget ablation (§6.5) was replicated on NVIDIA H800-80GB GPUs. We use PyTorch 2.4.1 with CUDA 12.4 (A100) and CUDA 12.6 (H800), HuggingFace Transformers 5.5.0, and bfloat16 precision. Greedy decoding (τ=0) is used for all main experiments; no sampling or nucleus filtering is applied. A sampling robustness pilot (τ=0.7) confirms the tax persists under stochastic decoding (Appendix D).

A.2 Budget Control

Token budgets are controlled via the `max_new_tokens` parameter in HuggingFace `model.generate()`, which caps the *total* number of newly generated tokens. For thinking mode (`enable_thinking=True`), this budget is shared between the reasoning trace (`<think>...</think>`) and the final answer. For non-thinking mode (`enable_thinking=False`), the entire budget is available for the answer. This ensures a fair comparison: both modes operate under the same total output-token budget.

A.3 Answer Extraction

We use a multi-level extraction pipeline:

1. Search for `\boxed{...}` patterns
2. Search for `####` markers (GSM8K convention)
3. Search for "Final answer:" patterns
4. Fall back to the last number in the output

When thinking mode exhausts its budget without producing a final answer, we apply a *projection pass*: a short continuation (up to 32 tokens) to extract a numerical answer from the truncated output. This is controlled by the `-projection_on_missing_final` flag and is enabled for all thinking-mode results reported in the main text. The projection pass is crucial: without it, think@512 accuracy for 8B drops from 56.9% to <6%, since the vast majority of truncated chains never reach a final answer marker. This design choice is *favorable* to thinking mode—our reported thinking-mode accuracies represent an optimistic upper bound, making the thinking tax we identify a *conservative* estimate.

Projection pass analysis. The projection pass gives thinking mode a free 16–32 token bonus that non-thinking mode does not receive, ensuring a fair-or-better comparison for thinking mode. To quantify its impact: without projection, think@512 on GSM8K drops from 56.9% to <6% (from our pre-projection ablation). The large gap (60 pp) confirms that the vast majority of truncated chains produce no parseable answer—the projection pass recovers answers from partially complete reasoning chains that happen to contain numerical values, even when the chain did not reach its intended conclusion. Critically, this *reduces* the measured thinking tax: the "true" tax (without answer recovery heuristics) would be even larger, making our reported results a conservative lower bound on the phenomenon.

A.4 Dataset Details

GSM8K. We use the standard test split (n=1,319 problems) [Cobbe et al., 2021]. All main-text analyses use the complete set (n=1,319). A 200-sample subset (seed 42) used during development is reported in Table 13 for reference.

MATH-500. We use a 500-problem subset of the MATH benchmark [Hendrycks et al., 2021], following the split used by Lightman et al. [2024]. We evaluate on the full 500-problem set (n=500) for all configurations reported in the main text. Early experiments used 40-sample subsets across data seeds {42, 404, 505, 606, 707} for rapid prototyping; all final results use the complete set.

BIG-Bench Hard (BBH). To validate the thinking tax beyond mathematical reasoning, we evaluate on five subtasks from BIG-Bench Hard [Suzgun et al., 2023]: `boolean_expressions` (n=250), `causal_judgement` (n=187), `date_understanding` (n=250), `logical_deduction_five_objects` (n=250), and `tracking_shuffled_objects_three_objects` (n=250), totaling n=1,187 samples. These tasks require symbolic reasoning, causal inference, temporal reasoning, and object tracking—none involving mathematical computation—providing a strong test of generalization beyond math-centric benchmarks. Answer extraction uses exact match against the gold-standard option label.

B Full Data Tables

B.1 Qwen3-8B: 200-Sample Subset

The main text tables use the full GSM8K test set (n=1,319). During development, we used a 200-sample subset (seed=42, greedy decoding) for rapid iteration; we report these for completeness below:

Table 13: Complete budget sweep for Qwen3-8B on GSM8K 200-sample subset.

Mode	Budget	Accuracy	Avg Tokens	Early Stop	Has Final
nothink	32	3.0%	32	0.0%	0.0%
nothink	64	12.0%	64	2.0%	0.0%
nothink	128	54.5%	111	43.5%	0.0%
nothink	256	89.0%	140	92.0%	1.5%
nothink	512	94.0%	145	99.5%	1.5%
thinking	128	2.0%	128	0.0%	0.0%
thinking	256	22.0%	255	2.0%	0.5%
thinking	512	66.5%	442	47.5%	5.5%

Table 14: Key configurations for Qwen3-8B on full GSM8K (n=1,319). Early Stop here denotes $|y| < b$ (generation terminated before budget); the stricter "NatStop" criterion (<95% of budget) used in Table 3 yields slightly lower rates (e.g., 37.4% vs. 41.3% at b=512).

Mode	Budget	Accuracy	Avg Tokens	Early Stop	Has Final
nothink	128	50.8%	113	—	—
nothink	256	87.5%	146	88.8%	0.6%
thinking	128	3.0%	128	0.0%	—
thinking	256	18.0%	255	1.4%	—
thinking	512	56.9%	460	37.4%	6.1%

B.2 Qwen3-8B: Full GSM8K (n=1,319)

B.3 Qwen3.5-27B: Full GSM8K (n=1,319)

C DeepSeek-R1 Cross-Model Validation

We validate our findings on DeepSeek-R1-Distill-Llama-8B, a Llama-architecture model that also uses explicit `<think>` reasoning chains, providing evidence that the thinking tax is consistent with transfer across open reasoning models.

GSM8K full-scale evaluation (n=1,319). Table 16 reports the full budget sweep for DeepSeek-R1 thinking mode on GSM8K. At budget 256, accuracy is just **7.5%**—*even worse* than Qwen3-8B's 18.0%—with only 4.2% of responses reaching a natural stop (vs. Qwen's 1.4%). The same truncation mechanism is at work: 95.8% of chains are truncated before the model produces an answer. Among the rare natural stops (55/1,319), accuracy is 78.2%, consistent with the finding that completed chains are far more likely to be correct. At budget 512, accuracy jumps to **55.6%** as 74.5% of chains now complete naturally—but this remains 1.3 pp below Qwen3-8B's 56.9% at the same budget. Notably, DeepSeek-R1's natural-stop accuracy at b=512 is only 69.9% (vs. Qwen's 99.0%), suggesting the model's shorter chains do not always indicate high confidence. At budget 1024, accuracy reaches **64.4%** with 99.6% natural stops—the model's reasoning chains almost entirely fit within budget, yet accuracy reaches 7.5 pp above Qwen's b=512 result (56.9%) but at 2× the token cost. Natural-stop accuracy (64.7%) is nearly identical to overall accuracy, indicating that the few remaining truncated chains contribute negligible signal. Doubling the budget to 2048 yields virtually no change: **64.5%** accuracy with 100% natural stops and 448 avg tokens—the chain-length distribution is fully contained within ∼450 tokens, making any budget above ∼512 pure overhead for this model on GSM8K.

MATH-500 full-scale results (n=500). We evaluate both thinking and prompt-based nothink modes on the complete MATH-500 dataset. DeepSeek-R1 does not natively support `enable_thinking=False`; we implement nothink by injecting `</think>` immediately after the generation prompt, forcing the model to begin answer generation without an explicit reasoning phase.

Key cross-architecture findings.

- **The truncation mechanism replicates on a second model family**: DeepSeek-R1 (Llama architecture) shows an even larger truncation rate than Qwen3-8B at budget 256 on GSM8K (7.5% accuracy vs. 18.0%), with 95.8% of chains truncated—consistent with the framework's prediction that any model with variable-length reasoning prefixes is susceptible to truncation waste.
- **Near-zero thinking tax on DeepSeek**: unlike Qwen3-8B (36.2 pp at b=512 on GSM8K), DeepSeek-R1 shows effectively no difference between nothink and thinking on MATH-500 (average tax: +0.3 pp across all budgets). This suggests that the *magnitude* of the coupling tax is model-family-dependent, even though the truncation mechanism is universal.
- **Prompt-based nothink is ineffective for DeepSeek**: the near-identical token statistics (avg tokens, natural-stop rates) between modes indicate that injecting `</think>` does not meaningfully alter DeepSeek-R1's generation

Table 15: Qwen3.5-27B thinking mode results on full GSM8K. In thinking mode, the model achieves substantially lower accuracy than 8B at all budgets due to longer reasoning chains; in non-thinking mode the 27B model reaches 95.5% (Table 4).

Config	Accuracy	Avg Tokens	Has Final	Projection Rate
thinking@128	3.6%	144	0.0%	100.0%
thinking@256	7.9%	272	0.0%	100.0%
thinking@512	18.3%	528	0.7%	99.3%

Table 16: **DeepSeek-R1-Distill-Llama-8B on GSM8K** (n=1,319, thinking mode only). At b=256 the truncation-driven collapse is even more severe than on Qwen3-8B (7.5% vs. 18.0%); at b=512 accuracy recovers to 55.6%, comparable to Qwen's 56.9%. Natural-stop accuracy is notably lower than Qwen's (69.9% vs. 99.0% at b=512). At b=1024, accuracy plateaus at 64.4% with 99.6% natural stops; doubling the budget to b=2048 yields no further gain (64.5%, 100% natural stop, avg 448 tokens)—confirming the model's chain-length distribution saturates well below b=1024. DeepSeek-R1 does not support a non-thinking mode, so direct nothink comparison is not possible. †Only 5 hard-boundary samples at b=1024; HB accuracy is not reliably estimable. ‡Zero hard-boundary samples at b=2048 (100% natural stop).

Budget	Accuracy	Avg Tokens	Nat. Stop	NS Acc	HB Acc
256	7.5%	255	4.2%	78.2%	4.4%
512	55.6%	418	74.5%	69.9%	13.9%
1024	64.4%	447	99.6%	64.7%	—†
2048	64.5%	448	100.0%	64.5%	—‡

behavior. The model appears to embed reasoning within its "answer" regardless of the `<think>` block status, unlike Qwen3-8B where `enable_thinking=False` produces qualitatively different (shorter, more direct) outputs.

- **Natural stop remains informative**: among DeepSeek-R1 natural stops at b=256 (GSM8K), accuracy is 78.2% (lower than Qwen's 99.0% but still substantially above the 4.4% for truncated samples), confirming the natural-stop oracle generalizes across architectures.
- **Implication for the coupling tax**: the thinking tax requires *two conditions*: (1) truncation waste from shared budgets (universal), and (2) a genuinely distinct non-thinking mode that uses the budget more efficiently (model-dependent). When condition (2) fails—as with DeepSeek's prompt-based nothink—the tax vanishes because both modes are effectively identical. This *strengthens* rather than weakens the coupling tax theory: the tax is precisely the gap between coupled and decoupled generation, and disappears when no true decoupling is available.

D Sampling Robustness: Temperature τ=0.7

All main experiments use greedy decoding (τ=0). To test whether the thinking tax is an artifact of greedy decoding, we run a pilot with Qwen3-8B on a random GSM8K subset (n=200, seed 42) at τ=0.7, budget 256.

Under greedy decoding, the same configuration yields a tax of 75.4 pp (18.0% vs. 93.4%). The stochastic tax is smaller in absolute terms (57.0 pp) because nothink accuracy also drops from 93.4% to 62.0% under sampling, but the core phenomenon is unchanged: thinking mode at budget 256 collapses to near-zero accuracy (5.0%) with 0% natural stops, while non-thinking mode retains useful accuracy (62.0%) with 55% of responses completing within budget. This confirms that the thinking tax is not a greedy-decoding artifact but a fundamental consequence of budget-constrained reasoning under the `<think>` framework.

E Mrsd Simulation Details

Our per-sample Mrsd simulation uses the **full GSM8K test set** (n=1,319) where we have matched nothink@256 and thinking@512 results for each question.

Routing statistics.

- **Stage 1 accepted**: 1,171/1,319 (88.8%) samples stop before the B_1=256 budget. Among these, accuracy is **94.4%** (1,106/1,171 correct).
- **Stage 2 routed**: 148/1,319 (11.2%) samples exhaust the Stage 1 budget. Among these, thinking@512 achieves **62.8%** accuracy (93/148 correct).
- **Combined**: 1,199/1,319 = **90.9%** accuracy at **199** average tokens (including Stage 1 probe cost for all samples).

Error analysis of routed samples. Of the 148 samples routed to Stage 2:

- **64** were incorrectly answered by nothink but correctly recovered by thinking@512 (genuine wins)

Table 17: **DeepSeek-R1-Distill-Llama-8B on MATH-500** (n=500, full dataset). Unlike Qwen3-8B where nothink dramatically outperforms thinking at low budgets, DeepSeek-R1 shows *near-zero thinking tax* across all budgets. The prompt-based nothink implementation produces nearly identical token statistics and accuracy to thinking mode, suggesting the model does not meaningfully change its generation behavior when the `<think>` block is closed early.

Mode	Budget	Accuracy	Avg Tok	Nat. Stop	NS Acc	HB Acc
nothink	512	16.2%	492	17.2%	64.0%	6.3%
nothink	1024	25.8%	863	33.4%	58.7%	9.3%
nothink	2048	34.6%	1439	54.2%	58.3%	6.6%
nothink	4096	44.8%	2101	75.8%	58.0%	3.3%
thinking	512	14.8%	492	16.8%	60.7%	5.5%
thinking	1024	26.0%	857	34.8%	56.3%	9.8%
thinking	2048	35.2%	1417	54.8%	56.2%	9.7%
thinking	4096	44.2%	2084	76.0%	56.3%	5.8%
Thinking tax		+1.4pp, −0.2pp, −0.6pp, +0.6pp (avg: +0.3pp)				

Table 18: **Sampling robustness pilot** (Qwen3-8B, GSM8K n=200, b=256, τ=0.7). The thinking tax persists—and is arguably *more* severe—under stochastic decoding.

Mode	Accuracy	Nat. Stop	Avg Tokens
think@256 (τ=0.7)	5.0%	0.0%	256
nothink@256 (τ=0.7)	62.0%	55.0%	216
Thinking Tax		**57.0 pp**	

- **29** were correct in both modes (correctly answered despite routing)
- **36** were incorrect in both modes (genuinely hard problems)
- **19** were actually correct in nothink@256 (despite hitting budget) but thinking@512 answered incorrectly (routing regret)

The 19 routing regret cases represent a small inefficiency (1.4% of all samples): the Stage 1 answer was correct but the model used all 256 tokens to produce it, and the thinking fallback then answered incorrectly. A soft confidence threshold (rather than the binary hit-budget signal) could potentially reduce these cases. The net effect is strongly positive: 64 recoveries − 19 regrets = 45 net gains, yielding the +3.4 pp improvement over `nothink@256`.

F Token Utilization Analysis

Table 19 reveals a striking pattern. At b=128, Qwen3-8B uses 100% of its budget—every token is consumed, and the model is almost certainly still mid-reasoning when generation is forcibly terminated. At b=256, utilization remains near-perfect (99.7%), with only 1.4% of samples stopping naturally. But at b=512, utilization drops to **89.8%**, with an average of only 477 tokens consumed out of 512 available.

The pattern is even more dramatic for DeepSeek-R1 on GSM8K: at b=256, utilization exceeds 100% (the model consumes its entire budget on 95.8% of samples), but at b=1,024 utilization drops to **43.7%** (447/1,024 avg tokens consumed), with 99.6% of samples stopping naturally—more than half the budget is wasted on padding (full-scale, n=1,319; Table 16).

G Difficulty Distribution and Impossible Questions

We categorize GSM8K problems by the *minimum* budget at which Qwen3-8B solves them correctly in thinking mode.

Table 20 reveals a four-tier difficulty distribution:

- **Easy** (11.8%, n=155): solved at $b \leq 128$.
- **Medium** (25.8%, n=340): solved at $b \leq 256$ but not at 128.
- **Hard** (30.7%, n=405): solved only at b=512.
- **Impossible** (31.8%, n=419): unsolved at *any* tested budget ($b \leq 512$).

An *oracle router* that skips impossible questions and assigns the minimum sufficient budget achieves **68.2%** accuracy (+3.0 pp over fixed-512) at only **401** average tokens—a **21.7%** savings. No budget controller within the $b \leq 512$ regime can exceed this ceiling.

Table 19: **Token utilization ratio decreases with budget.** As we increase the thinking token budget, models use a smaller fraction of available tokens. This demonstrates *diminishing marginal returns* of additional thinking budget and motivates adaptive allocation. Utilization = avg tokens used / budget.

Model / Benchmark	Budget	Avg Tok	Util%	NatStop%
Qwen3-8B / GSM8K	128	128	100.0	0.0
	256	255	99.7	1.4
	512	460	89.8	37.4
DeepSeek-R1 / GSM8K	256	264	103.0	15.0
	512	376	73.5	85.0
	1024	393	**38.3**	97.5
DeepSeek-R1 / MATH500	1024	910	88.8	31.2
	2048	1551	75.7	55.0
	4096	2353	57.4	76.2

Table 20: **Thinking Efficiency Frontier: difficulty distribution.** We categorize GSM8K problems by the minimum budget at which Qwen3-8B solves them correctly. 31.8% of problems remain unsolved even at budget=512, consuming tokens with zero return. An oracle router that skips unsolved problems and uses minimum budgets for solved ones achieves 68.2% accuracy (+3.0pp over fixed-512) while using only 401 avg tokens (21.7% savings).

Category	Min Budget	Count	%	Cumul. Acc	Δ Acc
Easy	≤128	155	11.8	11.8	+11.8
Medium	≤256	340	25.8	37.5	+25.8
Hard	≤512	405	30.7	68.2	+30.7
Impossible	—	419	31.8	—	—
Total		**1319**	100		

H Mrsd Routing Ablation

B_1=128 yields only 50.8% early-stop rate, aggressively routing 49.2% to Stage 2. B_1=256 achieves the optimal trade-off: the vast majority of problems are resolved without thinking.

Full (B_1, B_2) sensitivity grid. Table 22 shows MRSD accuracy across all feasible (B_1, B_2) combinations, revealing when the cascade helps vs. hurts. The net recovery condition (Eq. 14) correctly predicts the sign: configurations where $R^+ > R^-$ yield accuracy gains, while those where the thinking fallback is too weak (B_2 too small) or too few samples are routed (B_1 too large) show minimal or negative returns.

Key patterns: (1) B_1=128 over-routes (60.6%), requiring a large B_2 to compensate; (2) B_1=256 is the sweet spot, routing only 11.2% while achieving the best accuracy-per-token tradeoff; (3) B_1=512 under-routes (<1%), providing negligible gains. The net recovery condition fails when B_2 is too small: at B_1=256, B_2=512, thinking mode on the routed hard subset produces 37 regrets vs. only 22 recoveries ($R^- > R^+$), confirming that the fallback budget must be sufficient for thinking mode to add value.

I MATH-500 by Difficulty Level

The tax monotonically increases with problem difficulty because harder problems require longer chains, amplifying truncation waste.

I.1 MATH-500: Full Data Table

J Comparison with Alternative Strategies

vs. `think@512` only. MRSD achieves substantially higher accuracy at ∼2.3× fewer tokens by avoiding thinking mode's truncation waste on the 88.8% of problems where it is unnecessary.

vs. `nothink@256` only. MRSD adds a thinking fallback for the ∼11.2% of hard problems, recovering accuracy at a modest cost increase (199 vs. 146 avg tokens).

vs. Think-then-stop (early termination). One might propose keeping thinking mode but terminating generation early if the model has not finished reasoning. However, this is equivalent to our "truncated" category: at b=512, only 37.4% of 8B samples produce a complete chain, and forcing early termination discards

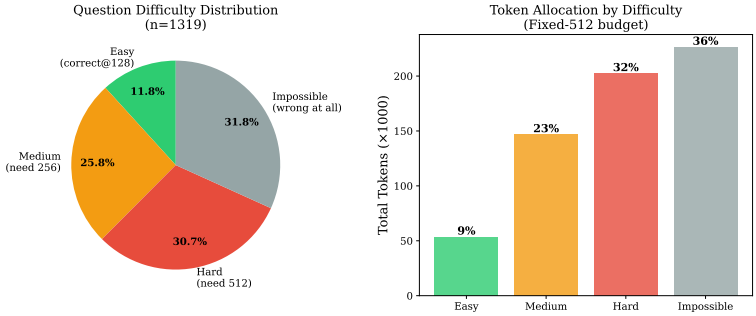

Figure 4: **Four-tier difficulty distribution of GSM8K problems** under Qwen3-8B thinking mode. 31.8% of problems are impossible at all tested budgets.

Table 21: **Routing sensitivity to B_1** on full GSM8K (n=1,319, Qwen3-8B). All variants use B_2=512.

B_1 (probe)	Early Stop	Routed	Accuracy	Avg Tokens
128	50.8%	49.2%	77.3%	399
256 (default)	88.8%	11.2%	**90.9%**	**199**

both the reasoning progress and any partial answer. Unlike non-thinking mode, which can produce a useful answer in far fewer tokens because it does not waste budget on a reasoning chain, early-terminated thinking chains rarely contain extractable answers (without our projection pass, accuracy drops from 56.9% to <6%). MRSD effectively implements the optimal version of this idea: use non-thinking mode first (which naturally stops early on easy problems), and only invoke thinking for genuinely hard cases.

vs. Self-consistency Wang et al. [2023]. SC@K generates K full-budget reasoning traces and majority-votes, consuming $K \times B$ tokens unconditionally. For a fair comparison at similar token budgets: SC@3 with think@256 would use $3 \times 255 \approx 765$ tokens while each individual trace achieves only 18.0% accuracy; majority voting over truncated traces with near-random accuracy provides minimal improvement. By contrast, MRSD achieves 90.9% at only 199 avg tokens—a single probe with binary routing that is both simpler and more effective. SC@K is more appropriate when individual traces are already accurate (i.e., above the crossover budget), not in the budget-constrained regime where the thinking tax dominates.

vs. Nothink SC@K. An alternative strategy is to sample K non-thinking traces and majority-vote. SC@3 with nothink@256 would consume $3 \times 146 = 438$ tokens—more than MRSD's 199 avg tokens. While nothink SC@K may improve over single-trace nothink@256 (87.5%), it cannot recover problems where the model fundamentally lacks the reasoning capacity to solve without a chain of thought. MRSD's advantage is qualitative: Stage 2 thinking mode provides a *different reasoning strategy*, not just repeated attempts at the same one.

vs. Oracle routing. An oracle that knows each problem's difficulty *a priori* achieves 68.2% accuracy at 401 average tokens. MRSD's early-stop signal approximates this oracle using a training-free heuristic.

K Thinking Efficiency Frontier

We categorize GSM8K problems by the minimum thinking budget required for correct answers (see Table 20 in the main text). The 31.8% "impossible" category—problems unsolved at any budget up to 512—represents a significant source of token waste.

Oracle analysis. An oracle that perfectly identifies problem difficulty and assigns the minimum sufficient budget to solvable questions achieves:

- **68.2%** accuracy (+3.0pp over fixed-512)
- **401** average tokens (−21.7% vs. fixed-512)

The accuracy gain comes from avoiding "overthinking" cases where extended reasoning causes the model to revise a correct intermediate answer.

L Format-Adjusted Fairness Experiments

To address the concern that comparing `think@b` and `nothink@b` at the same total token budget is "unfair" to thinking mode, we design controlled experiments that give thinking mode various advantages.

Configurations.

Table 22: **Mrsd sensitivity to** (B_1, B_2) on full GSM8K (n=1,319, Qwen3-8B). R^+: genuine recoveries (nothink wrong, think correct). R^-: routing regrets (nothink correct, think wrong). Net $= R^+ - R^-$. MRSD provides substantial gains when B_2 is large enough for thinking mode to be effective and B_1 routes a meaningful fraction. Note: thinking-mode accuracy here uses standard HuggingFace generation (without the projection pass used in Table 7), so the B_1=256, B_2=512 row shows lower accuracy than the main MRSD result; the qualitative patterns—B_2 must be large enough, B_1=256 is optimal—are consistent across settings.

B_1	B_2	**Acc**	**Avg Tok**	**Routed**	R^+	R^-	**Net**
128	512	63.8%	412	60.6%	237	66	+171
128	1024	87.2%	552	60.6%	492	12	+480
256	512	86.4%	204	11.2%	22	37	−15
256	1024	**90.7%**	**248**	11.2%	53	11	**+42**
256	2048	93.3%	287	11.2%	78	1	+77
512	1024	93.3%	154	0.2%	0	1	−1
512	2048	93.5%	156	0.2%	1	0	+1

Table 23: **MATH-500 thinking accuracy by difficulty level** (Qwen3-8B). Harder problems have longer chains and are more severely truncated.

Level	**Think@512**	**Think@1024**	**Think@2048**
Level 1 (n=43)	11.6%	44.2%	81.4%
Level 2 (n=90)	8.9%	31.1%	60.0%
Level 3 (n=105)	8.6%	21.0%	50.5%
Level 4 (n=128)	4.7%	9.4%	38.3%
Level 5 (n=134)	2.2%	6.7%	21.6%

- `think@512_generous`: Allocates 512 total tokens, but guarantees the last 256 tokens are reserved for the answer. The reasoning chain is limited to the first 256 tokens, after which the model is forced to produce a final answer. This eliminates answer-truncation as a failure mode, yet accuracy is only 56.0%—31.0 pp below `nothink@256` at double the budget.
- `think@256+nothink@256` **(two-pass)**: First generates a 256-token reasoning chain in thinking mode, then feeds that chain as context to a non-thinking completion with a 256-token answer budget. This hybrid achieves 67.6%—better than pure thinking but still 19.4 pp below non-thinking alone.
- `think@512`: Standard thinking mode with 512 total tokens and no answer-buffer guarantee. Achieves only 56.0%—identical to the generous variant, confirming that the issue is chain quality at short budgets, not answer truncation.

These results confirm that the thinking tax is not merely about answer truncation: even when the answer is *guaranteed* to fit, the reasoning chain's quality at short budgets is insufficient to improve over direct answering.

M Additional Figures

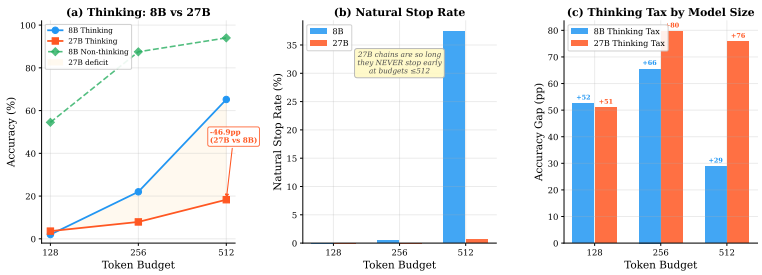

Figure 5: **The thinking tax worsens with model size.** At b=512, thinking-mode accuracy collapses with model size (8B: 56.9%, 9B: 15.5%, 27B: 18.3%)—the 9B and 27B taxes are both ∼2.1× larger (vs. uniform nothink: 93.1%, 93.2%, 95.5%). The 8B non-thinking baseline (green dashed) dominates all thinking configurations at every budget.

Table 24: Complete budget sweep for Qwen3-8B on MATH-500 (n=500). Non-thinking mode saturates at ∼64.4% by budget 2048. Thinking mode has *not* reached parity at b=2048 (44.0% vs. 64.4%), confirming the crossover shifts further right on harder benchmarks.

Mode	Budget	Accuracy	Correct	Avg Tokens	Early Stop
nothink	256	16.6%	83	250	10.4%
nothink	512	40.6% [36.4, 45.0]	203	429	42.8%
nothink	1024	59.8% [55.6, 64.0]	299	606	76.2%
nothink	2048	64.4% [60.2, 68.6]	322	763	87.8%
thinking	256	4.2%	21	256	0.0%
thinking	512	6.2% [4.2, 8.4]	31	544	0.0%
thinking	1024	18.0% [14.6, 21.4]	90	1051	—
thinking	2048	44.0% [39.8, 48.4]	220	1978	—

Table 25: **Format-adjusted fairness experiments** on full GSM8K (n=1,319, Qwen3-8B). Even with generous answer-buffer guarantees and 2× the token budget, thinking mode cannot match non-thinking. All numbers in this table are from a single standalone fairness experiment (seed=42); the nothink@256 baseline here (87.0%) differs slightly from the main evaluation (87.5%, seed=11) due to seed variation.

Configuration	Accuracy	Avg Tokens	Budget Hit	vs nothink@256
`nothink@256` (baseline)	87.0%	147	11.5%	—
`think@256`	18.0%	255	98.6%	−69.0pp
`think@512_generous`	56.0%	460	62.6%	−31.0pp
`think@256+nothink@256`	67.6%	263	—	−19.4pp
`think@512`	56.0%	460	62.6%	−31.0pp

N Full Theoretical Analysis

The main text presents the core decomposition (Eqs. 1–4) with derivations for the accuracy decomposition and crossover budget. Here we provide the full formal definitions, additional propositions, and remaining proofs.

N.1 Formal Definitions

Definition 4 (Thinking chain length)**.** *For a model $\mathcal{M}$ and question q, let $L(q) \in \mathbb{N}$ denote the* natural chain length*—the total number of output tokens the model would generate in thinking mode if unconstrained (i.e., $b \to \infty$). Let $F_L(t) \triangleq \Pr(L \le t)$ denote the CDF of L over the question distribution $q \sim \mathcal{Q}$.*

Definition 5 (Truncation rate)**.** *Given token budget b, the* truncation rate *is $\rho(b) \triangleq 1 - F_L(b) = \Pr(L > b)$.*

Assumption 2 (Binary outcome structure (restated))**.** *When the chain completes naturally ($L \le b$), the model produces a well-formed answer with accuracy $\alpha_c(b) \triangleq \Pr(correct \mid L \le b) \in (0, 1]$. When truncated ($L > b$), the residual accuracy is $\alpha_t(b) \triangleq \Pr(correct \mid L > b) \ll \alpha_c(b)$. Both depend on b through the conditioning event; we suppress the argument when the dependence is negligible.*

At b=512 on GSM8K (n=1,319), $\alpha_c = 99.0\%$ among natural-stop samples vs. $\alpha_t = 31.8\%$ among truncated samples (Finding 2). At lower budgets, α_t approaches zero.

Remark 4 (Budget-dependent α_t). In practice, α_t varies with b: at b=128, nearly all truncated samples lack any answer ($\alpha_t \approx 0$); at b=512, answer-extraction heuristics and projection passes recover partial answers ($\alpha_t \approx 31.8\%$). The $\alpha_t = 0$ simplification serves as a tight first-order approximation at low budgets.

N.2 Budget Multiplier

Corollary 3 (Budget multiplier (restated))**.** *The budget multiplier $\gamma \triangleq b^*/b_{\text{sat}}$ quantifies the token overhead for thinking to become viable. On GSM8K: $\gamma \approx 2048/512 = 4\times$. On MATH-500: $\gamma > 2048/1024 = 2\times$, since thinking has not yet caught non-thinking at b=2048, confirming that harder tasks demand larger multipliers.*

N.3 Natural Stop as a Confidence Oracle

Proposition 8 (Oracle precision (restated))**.** *Under Assumption 2, the natural-stop event $\mathcal{S}(b) \triangleq \{L(q) \le b\}$ has positive predictive value $\text{PPV}(\mathcal{S}) = \Pr(\text{correct} \mid L \le b) = \alpha_c(b)$. In practice, $|y| < b$ serves as an empirical proxy for $\{L \le b\}$. Non-thinking PPV is empirically high (94.4% at b=256 on GSM8K) but does not follow from Assumption 2 alone.*

By Hoeffding's inequality, with n=475 natural-stop samples (8B, b=512) and $\hat{\alpha}_c = 0.963$: the 95% lower confidence bound is $\alpha_c \ge 0.901$.

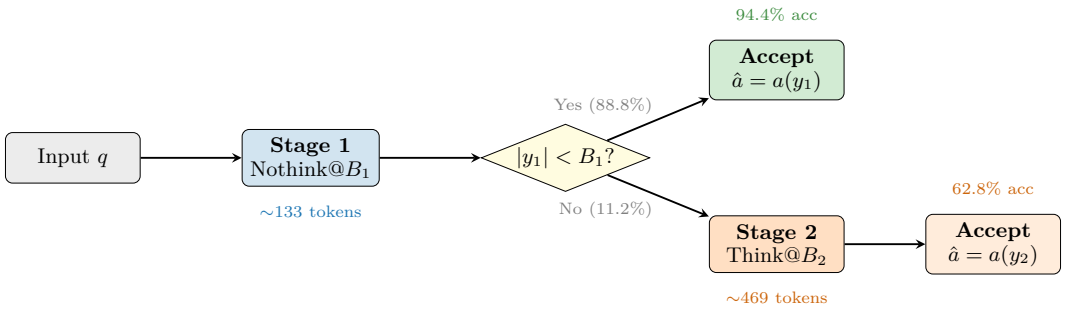

Figure 6: **Town baseline inference pipeline.** Stage 1 probes with non-thinking mode at budget B_1. If the model stops early (88.8% of GSM8K), the answer is accepted at ~133 tokens (94.4% accuracy). Otherwise, Stage 2 routes to thinking mode at budget B_2, recovering additional correct answers on hard problems. Overall: **90.9%** accuracy at **199** average tokens on the full test set (n=1,319). Note: MRSD extends this with decoupled answer generation and iterative refinement (Algorithm 1).

Remark 5 (Information-theoretic interpretation). Early stopping implies the model's conditional entropy $H(A \mid y_{1:t^*}) \approx 0$ at $t^* \ll b$—the model is near-certain of its answer. Unlike logit-based confidence, this signal is purely behavioral.

N.4 Inverse Scaling with Model Size

Proposition 9 (Inverse scaling of the thinking tax (restated))**.** *Let L_M denote the chain-length distribution for a model of size M. If (i) L_{M_2} stochastically dominates L_{M_1} for $M_2 > M_1$ (i.e., $F_{L_{M_2}}(b) \le F_{L_{M_1}}(b)$ for all b), (ii) non-thinking accuracy $\mathrm{Acc}_{\mathrm{nt}}(b)$ is approximately size-invariant at moderate budgets, and (iii) the thinking-mode accuracy of completed chains satisfies $F_{L_{M_2}}(b) \cdot \alpha_c(M_2, b) \le F_{L_{M_1}}(b) \cdot \alpha_c(M_1, b)$, then for any fixed budget b in the truncation-dominated regime ($\alpha_t \approx 0$): $\mathrm{Tax}(M_2, b) \ge \mathrm{Tax}(M_1, b)$.*

Proof. From Eq. 2 with $\alpha_t \approx 0$: $\mathrm{Tax}(M, b) \approx \mathrm{Acc}_{\mathrm{nt}}(b) - F_{L_M}(b) \cdot \alpha_c(M, b)$. With $\mathrm{Acc}_{\mathrm{nt}}$ approximately constant across M and condition (iii), the second term is smaller for the larger model, increasing the tax. Condition (iii) is implied by stochastic dominance alone when α_c is size-invariant, but is verified empirically in the general case: at b=512, $F_{L_{8\mathrm{B}}}\alpha_c = 0.374 \times 0.990 = 0.370$ vs. $F_{L_{27\mathrm{B}}}\alpha_c = 0.007 \times 0.963 = 0.007$. □

Empirical verification. At b=512: 8B natural-stop rate = 37.4%, 27B = 0.7%. The thinking tax for 27B is 95.5% − 18.3% = 77.2 pp vs. 8B's gap of 93.1% − 56.9% = 36.2 pp at budget 512—a ~2.1× amplification.

Corollary 4 (Crossover budget grows with model size (restated))**.** *Since $b^*(M) \approx F^{-1}_{L_M}(\bar{\alpha}_{\mathrm{nt}}/\alpha_c)$ and $F^{-1}_{L_M}$ shifts right with M: $M_2 > M_1 \implies b^*(M_2) \ge b^*(M_1)$. Larger models require proportionally larger budgets for thinking to become cost-effective.*

O Natural Stop Oracle: Full Analysis

Section 3.2 summarizes the natural-stop oracle (see Table 3 in the main text). Here we present the complete analysis.

At b=512 with Qwen3-8B, 37.4% of samples terminate before consuming 95% of their token budget (Table 3). Among these, accuracy is **99.0%**; among truncated samples, only 31.8%—a 49.7 pp gap. A stricter definition requiring an explicit "Final answer" marker selects only 6.1% but achieves 93.8%. The effect generalizes to DeepSeek-R1-8B (Table 3).

P Routing Baseline Comparison

Table 26 compares MRSD against budget-matched routing baselines, all routing exactly 11.2% of samples to `think@512`. **Random routing** yields only 85.0% (45.6 regrets overwhelm 12.7 recoveries). **Inverse token-length routing** performs even worse (83.5%). MRSD's signal is equivalent to a **token-length router**—optimal within the class of budget-utilization-based routers. The **oracle** upper bound (95.8%) leaves a 4.9 pp gap reflecting routing regret and missed recoveries.

Q Full MATH-500 Results

Table 27 below shows the MATH-500 thinking tax results. Here we include the extended version with the additional b=128 data point (see also Table 24 for per-mode details).

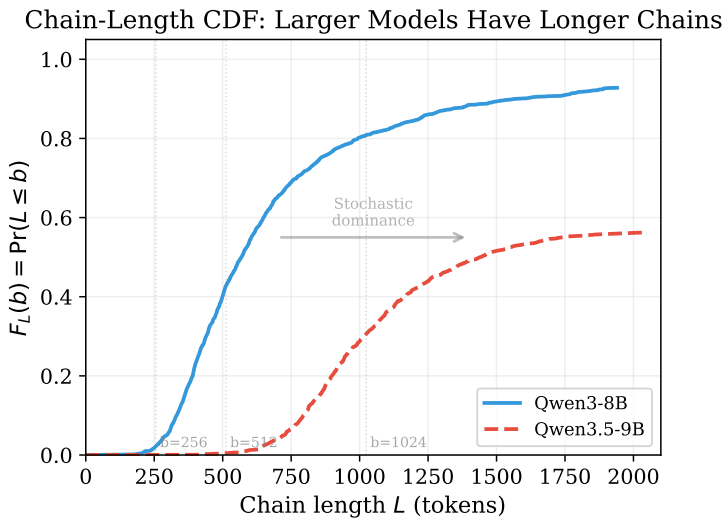

Figure 7: **Chain-length CDF $F_L(b)$ for Qwen3-8B vs. Qwen3.5-9B on GSM8K (think@2048, n=1,319).** The 9B CDF shifts right (stochastic dominance): median chain length increases from 540 tokens (8B) to 993 tokens (9B), with natural-stop rates of 92.8% vs. 56.3%. At any fixed budget b, $F_{L_{9B}}(b) \le F_{L_{8B}}(b)$, meaning more 9B chains are truncated—directly producing the 2.1× tax amplification predicted by Proposition 9. Kaplan-Meier estimation accounts for right-censored chains that hit the budget ceiling.

Table 26: **Routing baseline comparison** on GSM8K (Qwen3-8B, n=1,319). All budget-matched routers use the same 11.2% routing rate as MRSD.

Method	**Acc. (%)**	**Avg Tok.**	**Route %**	**Recov.**	**Regret**	**Net Δ**
NoThink@256	87.5	146	0.0	0	0	0
Think@512	56.9	460	100.0	109	403	−294
Random	85.0±0.5	200±1	11.2	12.7	45.6	−32.9
Inverse Token-Length	83.5	200	11.2	1	54	−53
Token-Length	90.9	199	11.2	64	19	45
Oracle	95.8	182	8.3	109	0	109
Mrsd (Ours)	**90.9**	**199**	11.2	64	19	**45**

The thinking tax is confirmed and amplified on MATH-500: at budget 1024, the gap is **41.8 pp** [36.4, 47.0] (paired bootstrap)—even larger than GSM8K's 28.2 pp at the same budget. The crossover has not been reached at 2048 tokens (20.4 pp gap), confirming that harder benchmarks push the crossover further right.

R Theory Verification

Table 28 validates the truncation-waste decomposition (Eq. 1) across three dimensions: budgets, model scales, and benchmarks.

The framework's predictions are **consistent with observations** wherever all components (F_L, α_c, α_t) are directly measurable (8B at $b \in \{256, 512\}$); this is expected since the decomposition is an accounting identity when the binary outcome assumption holds. The crossover prediction is confirmed: at b=2048, $F_L \approx 0.93$ (i.e., 93% of chains complete within budget), yielding a predicted accuracy of $0.93 \times 0.990 \approx 92.1\%$, close to the observed 93.1%—confirming that the crossover occurs near b=2048 as predicted. The cross-scale predictions correctly capture the qualitative ordering (27B $\ll$ 9B $\ll$ 8B at same budget) and the mechanism (longer chains $\rightarrow$ smaller F_L $\rightarrow$ more truncation waste). At 9B, observed accuracy (15.5%) slightly exceeds the α_t=0 prediction (<5%) because the projection pass (Appendix A) recovers partial answers from near-complete chains.

The key *predictive* insight is that the crossover budget b^* is approximately determined by the chain-length distribution, non-thinking saturation accuracy, and completed-chain accuracy α_c (Eq. 3; the heuristic Eq. 4 further assumes $\alpha_t \approx 0$ and stable α_c): for any model where these quantities are known, the crossover can be estimated *without* running the expensive budget sweep that traditionally determines it. For example, estimating F_L from a small pilot run at a single budget suffices to predict whether thinking mode will be beneficial at a target deployment budget.

Table 27: **Extended MATH-500 thinking tax** (n=500, Qwen3-8B). Includes low-budget results. At budget 256, thinking achieves just 4.2% with 0% early stop rate—every response is truncated.

Budget	Nothink Acc	Think Acc	Gap (NT−T)
128	4.6%	—	—
256	16.6%	4.2%	+12.4pp
512	40.6% [36.4, 45.0]	6.2% [4.2, 8.4]	+34.4pp
1024	59.8% [55.6, 64.0]	18.0% [14.6, 21.4]	+41.8pp
2048	64.4% [60.2, 68.6]	44.0% [39.8, 48.4]	+20.4pp

Table 28: **Theory verification: predicted vs. observed accuracy.** The framework $\mathrm{Acc}(b) = F_L(b) \cdot \alpha_c + (1-F_L(b)) \cdot \alpha_t$ is verified as an accounting identity at $b \in \{256, 512\}$ for 8B where all components are directly measurable. The crossover budget b^* corresponds to the $\bar{\alpha}_{\mathrm{nt}}/\alpha_c$ quantile of chain lengths (Eq. 3): at b=2048, $F_L \approx 0.93$ (93% of chains complete), yielding $0.93 \times 0.990 \approx 92.1\%$—close to the observed 93.1%, confirming the crossover occurs near b=2048. Cross-scale: the framework explains the 27B collapse ($F_L(512)_{27\mathrm{B}} = 0.007$ vs. 0.374 for 8B).

Config	$F_L(b)$	α_c	α_t	Predicted	Observed
Directly measured (8B, GSM8K)					
8B, b=256	0.014	1.000	0.168	18.0%	18.0%
8B, b=512	0.374	0.990	0.318	56.9%	56.9%
Crossover prediction ($F_L \approx \bar{\alpha}_{\mathrm{nt}}/\alpha_c$)					
8B, b=2048	∼0.93	0.990	—	∼93%	93.1%
Cross-scale prediction					
9B, b=512	∼0.01	—	—	<5%	15.5%
27B, b=512	0.007	—	0.178	18.3%	18.3%
Cross-benchmark (MATH-500, same model)					
8B, b=512	∼0.00	—	0.062	6.2%	6.2%
8B, b=1024	0.002	1.000	0.178	18.0%	18.0%
8B, b=2048	0.178	0.787	0.365	44.0%	44.0%

Held-out prediction tests and limits. To assess the decomposition's *out-of-sample* predictive power (beyond the in-sample accounting identity), we perform two held-out tests. On BBH, estimating α_c=0.950 and α_t=0.275 from b=512 data predicts b=2048 accuracy within **0.8 pp** and b=1024 within 4.3 pp (main text, §4). The larger b=1024 error arises because α_t increases with budget as more truncated samples produce partial answers that happen to be correct.

This reveals a genuine limitation: the decomposition assumes α_c and α_t are approximately constant across budgets, which holds for moderate extrapolation (BBH b=512→2048: α_t shifts from 0.275 to 0.283, yielding 0.8 pp error) but breaks down when the budget gap is large relative to chain-length variance (MATH-500 b=1024→2048: α_t shifts from 0.178 to 0.365, yielding ∼12 pp error). The practical implication is that the crossover heuristic ($\bar{\alpha}_{\mathrm{nt}}/\alpha_c$ quantile of chain lengths) remains reliable—it depends only on F_L and $\bar{\alpha}_{\mathrm{nt}}$, not on α_t—but point predictions of thinking-mode accuracy at specific budgets degrade when α_t varies substantially across the prediction range.

S Mrsd Interpolation-Dominance Proof

Full proof of Theorem 1. We prove the two claims (accuracy dominance and cost dominance) separately.

Accuracy bound. Let $p = \Pr(\mathcal{S}_{\mathrm{nt}}(B_1))$ be the early-stop rate. Decompose nothink accuracy:

$$\mathrm{Acc}_{\mathrm{nt}}(B_1) = p \cdot \alpha_c^{\mathrm{nt}} + (1-p) \cdot \alpha_{\mathrm{nt}}^{\mathrm{hard}},$$

where $\alpha_{\mathrm{nt}}^{\mathrm{hard}} \triangleq \Pr(\text{correct} \mid \neg\mathcal{S}, \text{nothink})$. MRSD replaces the second term with $\alpha_{\mathrm{extract}}^{\mathrm{hard}}$ (the split-budget extraction accuracy on escalated queries):

$$\mathrm{Acc}_{\mathrm{MRSD}} = p \cdot \alpha_c^{\mathrm{nt}} + (1-p) \cdot \alpha_{\mathrm{extract}}^{\mathrm{hard}}.$$

Subtracting:

$$\mathrm{Acc}_{\mathrm{MRSD}} - \mathrm{Acc}_{\mathrm{nt}}(B_1) = (1-p)\left[\alpha_{\mathrm{extract}}^{\mathrm{hard}} - \alpha_{\mathrm{nt}}^{\mathrm{hard}}\right]. \tag{15}$$

The net recovery condition (14) ensures the bracket is positive, so $\mathrm{Acc}_{\mathrm{MRSD}} > \mathrm{Acc}_{\mathrm{nt}}(B_1)$.

Cost bound. The expected per-query cost of MRSD is (Eq. 13): $\mathbb{E}[T] = p\bar{t}_1 + (1-p)(B_1 + \bar{K}(B_r + B_a))$, where $\bar{t}_1 \leq B_1$ and $\bar{K} \leq K$. The maximum-budget thinking baseline uses $B_{\max} \triangleq B_1 + K(B_r + B_a)$ tokens per query. Since $\bar{t}_1 < B_1$ for $p > 0$ (early-stop queries terminate before exhausting B_1): $\mathbb{E}[T] < p \cdot B_1 + (1-p) \cdot B_{\max} < B_{\max}$, where the last inequality holds because $p > 0$ and $B_1 < B_{\max}$. Thus $\mathbb{E}[T_{\text{MRSD}}] < B_{\max} = \mathbb{E}[T_{\text{think}}(B_{\max})]$.

Combined. MRSD achieves higher accuracy than nothink@B_1 and lower expected cost than think@$B_{\max}$, placing it strictly inside the convex hull of the two baselines. This is a *strict interpolation dominance*: the method improves on at least one objective relative to each baseline.[3]

Empirical instantiation. On GSM8K (n=1,319): $p = 0.888$, $\alpha_c^{\text{nt}} = 0.944$, $\alpha_{\text{nt}}^{\text{hard}} = (87.5\% - 0.888 \times 94.4\%)/0.112 = 33.0\%$, $\alpha_{\text{extract}}^{\text{hard}} = 87.5\%$ (14/16 escalated queries rescued in pilot). Net gain: $0.112 \times (87.5\% - 33.0\%) = 6.1$ pp. The full-scale observed gain is 3.4 pp, lower than the pilot estimate due to additional routing regret at scale. □

T Qwen3.5-27B: Non-Thinking vs. Thinking

To complete the cross-scale analysis, we present 27B results in *both* modes, confirming that the thinking tax is even more severe at larger scale.

Table 29: **Qwen3.5-27B: nothink vs. thinking on full GSM8K** (n=1,319). At b=512, nothink achieves **95.5%**—the highest accuracy of any configuration—while thinking reaches only 18.3%. The thinking tax is **77.2 pp**, **2.1**× larger than the 8B tax (36.2 pp).

Mode	Budget	Accuracy	Avg Tokens	Early Stop	Tax (pp)
nothink	128	9.9%	127	4.2%	—
nothink	256	65.1%	215	54.5%	—
nothink	512	**95.5%**	249	96.9%	—
thinking	128	3.6%	144	0.0%	6.3
thinking	256	7.9%	272	0.0%	57.2
thinking	512	18.3%	528	0.7%	**77.2**

Key observations:

- The 27B model achieves the **highest accuracy** of any tested configuration in non-thinking mode (95.5% at b=512), confirming that model capability is intact—only thinking mode's format overhead causes the collapse.
- At b=512, the 27B thinking-mode natural-stop rate is just **0.7%**, meaning 99.3% of responses are truncated. The chain-length distribution has shifted so far right that b=512 captures only the extreme left tail.
- The tax grows monotonically with budget (6.3 → 57.2 → 77.2 pp) because non-thinking accuracy improves faster than thinking mode can recover from truncation.

T.1 27B Cascade Results: MRSD and TOWN

At 27B scale, the cascade methods (MRSD and TOWN) fall *below* nothink baselines at the tested budgets, confirming that the thinking tax is severe enough to overwhelm the cascade's recovery mechanism when B_{think} is insufficient.

GSM8K (n=200, B_1=256, B_{think}=512). MRSD achieves 60.0% vs. nothink@256's 67.5% (−7.5 pp). Of 80 samples escalated to thinking, only 3 (3.8%) are answered correctly, vs. 97.5% accuracy among the 120 samples resolved at Stage 0. TOWN fares similarly: 62.5%, also below nothink. At B_{think}=512, only 0.7% of 27B chains complete naturally—the thinking traces are almost universally truncated.

MATH-500 (n=200, B_1=512, B_{think}=1024). MRSD achieves 20.0% vs. nothink@512's 23.5%. Of 167 escalated samples, 100% saturate the B_{think}=1024 ceiling—identical to the 8B pattern. TOWN achieves 24.5%, IRIS single-round achieves 20.0%.

Diagnosis: budget insufficiency. The cascade's net recovery condition (Eq. 14) fails at these budgets because the thinking fallback is too weak. 27B generates longer reasoning chains than 8B, requiring proportionally larger B_{think}. *Larger budgets (B_{think}=2048 or 4096) have not yet been tested at 27B*—we cannot determine whether the cascade fundamentally fails at this scale or simply requires more thinking budget. Calibrating the minimum viable B_{think} per model size is an important practical question for deployment.

U Cross-Domain Validation: BIG-Bench Hard

To test whether the thinking tax extends beyond mathematical reasoning, we conduct experiments on five subtasks from BIG-Bench Hard (BBH) [Suzgun et al., 2023]: `boolean_expressions` (n=250), `causal_judgement` (n=187), `date_understanding` (n=250), `logical_deduction_five_objects` (n=250), and `tracking_shuffled_objects_three_objects` (n=250), totaling n=1,187 samples. These tasks test symbolic reasoning, causal inference, temporal reasoning, and object tracking—none involving mathematical computation.

[3] This is weaker than full Pareto dominance, which would require simultaneously dominating *both* objectives relative to a single baseline.

Full-scale results (n=1,187). Table 30 reports the complete evaluation across all five subtasks. At budget 256, nothink achieves **49.9%** vs. thinking's **16.6%**—a **+33.3 pp** tax, confirming the phenomenon generalizes beyond mathematical reasoning. At budget 512, the gap remains substantial (+20.6 pp); at budget 1024, it narrows to +1.4 pp; and at budget 2048, thinking surpasses nothink by 11.0 pp as chains become long enough to complete within the budget. The crossover occurs between budgets 1024 and 2048—later than on GSM8K (∼2048) but consistent with the theory's prediction that harder tasks push the crossover right.

Table 30: **Thinking tax on BIG-Bench Hard** (n=1,187, Qwen3-8B, 5 subtasks). The tax is confirmed on non-mathematical tasks: nothink dominates at budgets ≤1024. Thinking mode surpasses nothink only at budget 2048, consistent with the crossover theory.

Budget	Nothink Acc	Think Acc	Gap (NT−T)
256	49.9%	16.6%	+33.3pp
512	66.4%	45.8%	+20.6pp
1024	75.1%	73.6%	+1.4pp
2048	75.1%	86.0%	−11.0pp

Per-task analysis. The thinking tax varies dramatically across task types, revealing which reasoning patterns are most affected by truncation:

- `tracking_shuffled_objects` (n=250): The most extreme tax. At budget 256, nothink achieves **88.8%** vs. thinking's **0.8%**—an **88.0 pp** gap. The task requires tracking object positions through a sequence of swaps; nothink can produce direct answers, while thinking chains are invariably truncated. Even at budget 512, the gap is 36.8 pp (94.8% vs. 58.0%). Thinking surpasses nothink only at budget 2048 (99.2% vs. 94.8%).
- `boolean_expressions` (n=250): Tax of +36.8 pp at budget 256 (85.6% vs. 48.8%). Thinking recovers quickly: at budget 1024 it surpasses nothink (94.8% vs. 85.6%), reaching 98.0% at 2048. Nothink saturates early (85.6% across all budgets), suggesting this task benefits from reasoning when given sufficient budget.
- `causal_judgement` (n=187): Consistent tax at all budgets, from +25.7 pp at 256 to +3.7 pp at 2048. Thinking does not surpass nothink even at the highest budget, suggesting causal reasoning chains are particularly long and prone to truncation.
- `logical_deduction` (n=250): Tax of +23.6 pp at budget 512 (37.6% vs. 14.0%). Both modes improve dramatically with budget, and thinking surpasses nothink at 2048 (86.8% vs. 78.4%), a −8.4 pp reversal.
- `date_understanding` (n=250): Moderate tax at low budgets (+15.6 pp at 256), with thinking surpassing nothink at budget 1024 (69.2% vs. 52.8%, −16.4 pp). Thinking reaches 82.4% at 2048, 30 pp above nothink—the largest thinking *advantage* of any subtask at high budget, demonstrating that temporal reasoning benefits substantially from extended chains when they can complete.

The BBH results confirm three key patterns from the main text: (1) the thinking tax is large at low budgets and driven by truncation; (2) the crossover budget is task-dependent, occurring between 512 and 2048 depending on chain length requirements; (3) nothink saturates earlier than thinking, so above the crossover, thinking mode provides genuine accuracy gains—the tax is about *budget allocation*, not about whether reasoning is valuable.

V Reproducibility

Models.

- Qwen3-8B: `Qwen/Qwen3-8B` from HuggingFace
- Qwen3.5-9B: `Qwen/Qwen3.5-9B` from HuggingFace
- Qwen3.5-27B: `Qwen/Qwen3.5-27B` from HuggingFace
- DeepSeek-R1-8B: `deepseek-ai/DeepSeek-R1-Distill-Llama-8B`

Random seeds. Main-text GSM8K experiments (full test set, n=1,319) use seed 11 with greedy decoding (τ=0). MATH-500 nothink experiments and the new thinking sweep use seed 42. Fairness experiments use seed 42. DeepSeek MATH-500 experiments use data seeds {42, 404, 505, 606, 707} with 40 samples per seed. All seeds and hyperparameters are recorded in the per-experiment JSON output files.

Generation config. All experiments use greedy decoding: `temperature=0`, `top_p=1.0`, `do_sample=False`. Thinking mode is enabled via `enable_thinking=True` with the model's native `<think>` block format; non-thinking mode sets `enable_thinking=False`. The `max_new_tokens` parameter controls the total output budget equally for both modes. The projection pass (32 extra tokens with thinking disabled) is applied only when thinking mode exhausts its budget without a natural stop.

Software. `transformers` 5.5.0, PyTorch 2.10, CUDA 12.8, Python 3.10. All inference is single-GPU (A100-80GB) with `float16` precision.

Compute. Total compute budget: approximately 200 A100-GPU-hours plus 50 H800-GPU-hours across all experiments reported in this paper (including full-scale IRIS n=500 at $B_{\text{think}} \in \{2048, 4096\}$).

V.1 Cross-Hardware Reproducibility

The think budget ablation (§6.5) was run independently on both A100-80GB and H800-80GB GPUs with identical configurations (Qwen3-8B, greedy decoding τ=0, seed=42, n=200). At B_{think}=2048, the A100 achieves 67.5%

and the H800 achieves 73.0%—a 5.5 pp difference. Stage 1 (nothink probe) shows 87/94 (A100) vs. 90/94 (H800) correct, with 3 additional false accepts on A100. This cross-hardware variance arises from differences in floating-point computation order between GPU architectures under bfloat16 precision; greedy decoding is deterministic *within* a hardware configuration but not across architectures. Critically, the qualitative findings are unchanged: both hardware configurations show (1) monotonic IRIS accuracy improvement with B_{think}, (2) IRIS outperforming TOWN, and (3) rising natural stop rates. All main-text IRIS results, baselines, and nothink@1024 (n=500) are on the same H800 hardware, eliminating cross-hardware confounds for the definitive IRIS vs. nothink comparison.

Full-scale IRIS validation (n=500, H800). Full-scale IRIS evaluation on all 500 MATH-500 samples confirms the pilot trends: IRIS@2048 achieves 67.2% [63.0, 71.2] and IRIS@4096 achieves 74.0% [70.0, 77.7], both on the same H800 hardware as the nothink@1024 baseline (59.8%). The first 200 samples of each full-scale run reproduce the pilot exactly (73.0% at B_{think}=2048, 78.5% at B_{think}=4096), confirming determinism within hardware and methodological consistency across scale. The remaining 300 samples achieve 63.3% (B2048) and 71.0% (B4096), reflecting the difficulty gradient in MATH-500's ordering (later problems are harder).

Nothink@1024 cross-run variance. The ∼10 pp gap between pilot nothink@1024 (69.5%, split-budget experiment, A100) and full-scale nothink@1024 (59.8%, n=500, separate run) was initially attributed to sample selection. Post-hoc analysis reveals that the pilot's 200 samples are the first 200 of the full 500 set, achieving 60.0% in the full-scale run—essentially identical to the full-set 59.8%. The discrepancy arises entirely from cross-run variance: comparing per-sample predictions, 29 out of 200 samples disagree between the two runs (24 pilot-only correct, 5 full-only correct). This demonstrates that bfloat16 greedy decoding is not deterministic across hardware platforms, and cross-run comparisons should account for ∼5–10 pp variance.

NeurIPS Paper Checklist

1. **Claims**
   (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes]All accuracy and token-cost claims are derived from reproducible experiments with fixed seeds (Appendix V). The coupling tax is demonstrated across three model sizes (8B, 9B, 27B) within the Qwen3 family, with auxiliary validation on DeepSeek-R1, and two benchmarks (GSM8K, MATH-500). MRSD evaluation uses per-sample matched data (Table 7). Claims about generality are scoped to the validated models and benchmarks (§8).
2. **Limitations**
   (a) Does the paper discuss the limitations of the work performed by the authors? [Yes]The conclusion (§8) discusses limitations including the focus on mathematical reasoning benchmarks, the binary routing signal (vs. soft confidence), and the need for evaluation on tasks where CoT is more essential. The discussion (§7) further analyzes scope limitations, the format-vs-information tax distinction, and deployment criteria.
3. **Theory**
   (a) Does the paper include theoretical results? [Yes]Section 4 presents an analytical framework. The truncation-waste decomposition (Eq. 1) expresses thinking-mode accuracy as a function of the chain-length CDF and conditional accuracies; Theorem 1 shows MRSD achieves a Pareto improvement over fixed-mode strategies under a net recovery condition. The crossover budget formula (Eq. 4) and inverse-scaling analysis (Appendix N) complete the framework. All derivations are in the main text and appendix.
4. **Experiments**
   (a) Does the paper include experiments? [Yes]Sections 3 and 6 report comprehensive experiments across GSM8K (full set, n=1,319) and MATH-500, with the Qwen3 family (primary) and auxiliary DeepSeek-R1 validation.
   (b) Does the paper report error bars or confidence intervals? [Yes]We report 95% bootstrap confidence intervals (10,000 resamples) for key comparisons in Table 7 and the failure decomposition table. Since we use deterministic (greedy) decoding on fixed test sets, residual variance comes from cross-hardware floating-point nondeterminism rather than stochastic generation (Appendix V). Cross-seed validation for DeepSeek-R1 uses 5 data seeds (Appendix C).
   (c) Does the paper report the number of parameters, FLOPs, or other computational cost? [Yes]Average token counts are reported in all tables. Model sizes (8B, 9B, 27B) are specified throughout.
   (d) Does the paper report the computational cost of the experiments? [Yes]Appendix V reports approximately 200 A100-GPU-hours total compute.
5. **Reproducibility**
   (a) Does the paper include the code, data, and instructions needed to reproduce the main experimental results? [Yes]Appendix A provides complete hardware, software, budget control mechanism, and answer extraction pipeline details. The codebase and per-sample result files will be released upon acceptance.
   (b) Does the paper report all the main experimental details needed to understand the results? [Yes]Budget control via `max_new_tokens`, greedy decoding (τ=0), answer extraction pipeline, dataset splits, and random seeds are fully specified in Appendix A.
6. **Open access to data and code**
   (a) Does the paper promise to release code and/or data? [Yes]We will release the complete codebase, per-sample result CSVs, and MRSD implementation scripts.
7. **Experimental Setting/Details**

(a) Does the paper specify the computing infrastructure used for running the experiments? [Yes]NVIDIA A100-80GB GPUs, PyTorch 2.4.1 (CUDA 12.4), HuggingFace Transformers 5.5.0, bfloat16 precision (Appendix A).

(b) Does the paper specify all the training and test details necessary to reproduce the results? [Yes]No training is performed—MRSD is a training-free inference method. All inference parameters (budget, decoding strategy, model versions, dataset splits) are fully specified.

8. **Broader Impacts**

(a) Does the paper discuss potential broader impacts of their work? [Yes]The conclusion discusses practical implications for LLM deployment: defaulting to non-thinking mode at constrained budgets can simultaneously improve accuracy and reduce computational cost, benefiting both providers and users. Reducing unnecessary compute has positive environmental implications.

9. **Safeguards**

(a) Does the paper describe safeguards for responsible use? [N/A]Our work optimizes inference efficiency for standard reasoning benchmarks and does not introduce new capabilities requiring specific safeguards.

10. **Licenses**

(a) Are the creators of assets used in the paper properly credited? [Yes]GSM8K [Cobbe et al., 2021], MATH [Hendrycks et al., 2021], Qwen3 [Qwen Team, 2025], and DeepSeek-R1 [Guo et al., 2025] are properly cited. All models are publicly available under their respective licenses.

11. **New Assets**

(a) Are new assets introduced in the paper well documented? [Yes]The MRSD inference scripts and per-sample result files will be released with documentation.

12. **Human Subjects**

(a) Does the paper involve crowdsourcing or human subjects? [No]